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571-272-7822

Paper 50
Entered: January 25, 2023

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

ILLUMINA, INC.,
Petitioner,

v.

RAVGEN, INC.,
Patent Owner.

IPR2021-01272
Patent 7,332,277 B2

Before ZHENYU YANG, TIMOTHY G. MAJORS, and DAVID COTTA, *Administrative Patent Judges*.

MAJORS, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining No Challenged Claims Unpatentable
35 U.S.C. § 318(a)
Granting Patent Owner's Motion for Entry of Protective Order and to Seal;
Granting Petitioner's Motion to Seal
37 C.F.R. §§ 42.14, 42.54

I. INTRODUCTION

Illumina, Inc. ("Petitioner" or "Illumina"),[1] on July 20, 2021, filed a Petition to institute *inter partes* review of claims 55–63, 66–69, 80–91, 94–96, 126–130, 132, and 133 of U.S. Patent No. 7,332,277 B2 (Ex. 1001, "the '277 patent"). Paper 1 ("Pet." or "Petition"). We instituted trial on January 26, 2022. Paper 14 ("Inst. Dec."). During trial, Ravgen, Inc. ("Patent Owner")[2] filed a Patent Owner Response. Paper 22 ("PO Resp."). Later filings include Petitioner's Reply (Paper 34 ("Pet. Reply")) and Patent Owner's Sur-Reply (Paper 44 ("PO Sur-Reply")). We also allowed Petitioner a limited Sur-Sur-Reply. Paper 46.[3] An oral hearing was held on October 26, 2022, and a transcript is entered in the record. Paper 49 ("Tr.").

Patent Owner also filed a motion for entry of a protective order and to seal (Paper 23), and Petitioner filed a motion to seal (Paper 31). Both motions were unopposed.

We have jurisdiction under 35 U.S.C. § 6(b). After considering the full record developed through trial, we determine that Petitioner has not proved by a preponderance of the evidence that the challenged claims are unpatentable. *See* 35 U.S.C. § 316(e). Our reasoning is explained below, and we issue this Final Written Decision under 35 U.S.C. § 318(a).

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[1] Petitioner identifies itself, Illumina Cambridge, Ltd., and Verinata Health, Inc., as the real parties-in-interest. Pet. 3.
[2] Patent Owner identifies itself as the real party-in-interest. Paper 4, 1.
[3] The permitted scope of this filing is detailed in Exhibit 3007 (*see* email to parties dated October 4, 2022, responding to Petitioner's request).

A. Related Patents & Proceedings

The ’277 patent issued February 19, 2008, from U.S. Patent Application No. 10/661,165 (“the ’165 Application”), filed September 11, 2003. Ex. 1001, codes (21), (22), (45). The ’165 Application is a continuation-in-part (“CIP”) of Application No. PCT/US03/06198 filed February 28, 2003, which claims priority to other applications including U.S. Provisional Application No. 60/378,354, filed May 8, 2002. *Id.* at codes (60), (63); *see also id.* at 1:7–25. The ’165 Application is also a CIP of Application No. PCT/US03/27308, filed August 29, 2003, and a CIP of U.S. Patent Application 10/093,618, filed March 11, 2002, which itself claims priority to U.S. Provisional Patent Application No. 60/360,232, filed on March 1, 2002. *Id.* at codes (60), (63); *see also id.* at 1:7–25.

Related U.S. Patent No. 7,727,720 (“the ’720 patent”), which also lists the ’165 Application in its priority chain, issued on June 1, 2010. *See* IPR2021-00791, Paper 20 at 2–3 (citing, as Exhibit 1001, the ’720 patent).

Petitioner identifies multiple lawsuits involving the ’277 patent. The lawsuits include: *Ravgen, Inc. v. Illumina, Inc.*, No. 20-1644 (D. Del.); *Ravgen, Inc. v. Natera, Inc. and NSTX, Inc.*, No. 1:20-cv-00692-ADA (W.D. Tex.); *Ravgen, Inc. v. Laboratory Corp. of America Holdings*, No. 6:20-cv-00969-ADA (W.D. Tex.); and *Ravgen, Inc. v. Quest Diagnostics*, No. 6:20-cv-00972-ADA (C.D. Cal.). Pet. 3 (identifying additional lawsuits against, e.g., PerkinElmer, Inc., and Myriad Genetics, Inc.); Paper 10.

Petitioner and Patent Owner also identify other matters involving the ’277 patent before the Patent Office. Pet. 3–4; Paper 4, 1–2; Paper 6, 1. Claims of the ’277 patent have been challenged in IPR2021-00788, -00789 and -00790 (all filed by Quest), IPR2021-00902 and -01054 (both filed by Labcorp), and IPR2021-01577 (filed by Streck, Inc.). Pet. 3–4; Paper 4, 1; Paper 6, 1. On October 19, 2021, we instituted trial in IPR2021-00788, and denied institution in IPR2021-00789 and IPR2021-00790. On November 5, 2021, we instituted trial in IPR2021-00902 and -01054 (covering, collectively, claims 55–63, 66–69, 80–96, and 127–133). We terminated IPR2021-00788 due to settlement (IPR2021-00788, Paper 71), and entered final written decisions in IPR2021-00902 and -01054 on November 1, 2022 (finding Labcorp had not shown the challenged claims were unpatentable). *See* IPR2021-00902 (Paper 55); IPR2021-01054 (Paper 51). IPR2021-01577 is ongoing. *Ex Parte* Reexamination Control No. 90/014,792 is also related to the ’277 patent. Paper 4, 2. That reexamination is stayed. *See* IPR2021-00902, Paper 24 (staying reexamination during pendency of the IPRs challenging the ’277 patent).

The related ’720 patent has also been challenged in several matters before the Office: IPR2021-00791 (Quest; terminated by settlement), -01026 (Labcorp; final written decision entered Dec. 12, 2022), and -01271 (Illumina; final written decision entered concurrently with this Decision); and *Ex Parte* Reexamination Control Nos. 90/014,703, and 90/014,869 (both stayed). IPR2021-00791, Paper 25.

B. Asserted Grounds of Unpatentability

Petitioner asserts four grounds of unpatentability in its Petition (Pet. 5–6), which grounds are identified in the table below:

Claims Challenged	35 U.S.C. §	Reference(s)/Basis
55–59, 61–63, 66–69, 80–89, 94, 96, 126–130	102(a), (e)[4]	Landes[5]
95	103(a)	Landes, Marx[6]
55–63, 66–69, 80–91, 94, 96, 126–130, 132, 133	103(a)	Lo,[7] Valli[8]
95	103(a)	Lo, Valli, Marx

Petitioner relies on the declarations of Dr. Brynn Levy, among other evidence. Ex. 1056; Ex. 1068. During trial, Patent Owner submitted and relies on a declaration from Dr. Brian Van Ness, Ph.D. (Ex. 2239), among

[4] The Leahy-Smith America Invents Act, Pub. L. No. 112-29, 125 Stat. 284 (2011) ("AIA"), amended 35 U.S.C. §§ 102 and 103. Based on the filing date of the '277 patent, we apply the pre-AIA versions of §§ 102 and 103.
[5] Landes et al., WO 03/062441 A1, published July 31, 2003 (Ex. 1003, "Landes").
[6] G. Marx et al., *Reducing white cells in platelet units*, 31:8 TRANSFUSION 743–747 (1991) (Ex. 1029, "Marx").
[7] Lo et al., WO 98/39474, published Sept. 11, 1998 (Ex. 1033, "Lo").
[8] V.E. Valli et al., *An Anticoagulant for Transport of Bovine Blood*, 21 CAN. VET. J. 252–257 (1980) (Ex. 1034, "Valli").

other evidence.[9] The deposition testimony of Drs. Levy (Exs. 2294, 2299) and Van Ness (Exs. 1067, 1069, and 1070[10]) is also of record.

C. Technology Overview and the '277 Patent

The '277 patent relates to non-invasive methods for sampling DNA and detection of genetic disorders in a fetus. Ex. 1001, 1:31–39. The '277 patent explains that a variety of invasive and non-invasive techniques are available for prenatal diagnosis, including amniocentesis, and analysis of fetal blood cells in maternal blood. *Id.* at 2:53–57. According to the patent, however, "techniques that are non-invasive are less specific, and the techniques with high specificity and high sensitivity are highly invasive." *Id.* at 2:57–60; *see also id.* at 3:33–37 (citing increased fetal mortality risk with amniocentesis, an invasive procedure).

We provide a brief overview of various components of blood, which is helpful in understanding the subject matter of the '277 patent and the asserted prior art. Blood is composed of plasma and other blood components suspended in plasma. Ex. 2239 ¶ 26. Major blood components

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[9] The parties submit additional witness testimony related to, for example, commercial success of products alleged to practice the claimed subject matter. *See, e.g.*, Exs. 2080 (Declaration of Paul Meyer) and 2242 (Declaration of Jeffrey Chalmers, Ph.D.).

[10] Exhibit 1069 is deposition testimony taken in IPR2021-01026, and Exhibit 1070 is deposition testimony taken in IPR2021-00788.

in plasma include red blood cells ("RBCs"), white blood cells ("WBCs") and platelets. *Id.*

Most DNA is typically found inside cells (within the cell membrane and nucleus); some DNA may, however, also be found outside the cells circulating freely in the plasma. *Id.*; Ex. 1056 ¶ 21 n.2. Extracellular circulating DNA is known as "cell-free DNA" ("cfDNA"). Ex. 2239 ¶ 26. WBCs, unlike RBCs, include an individual's cellular DNA, and when WBCs lyse (i.e., rupture of the cell), such lysis may release additional DNA into the circulating blood or a blood sample. Ex. 2239 ¶¶ 26–27; Ex. 1056 ¶ 23 (discussing lysis due to clotting when serum samples are prepared; "when the cells lyse, the nucleus and the cell break open, which releases the cell-free maternal DNA into the sample").

By the late 1990s, and prior to the '277 patent, researchers had discovered that pregnant women have cell-free *fetal* DNA ("cffDNA") along with maternal cfDNA circulating in maternal plasma. Ex. 2239 ¶ 27; Ex. 1056 ¶ 21. For example, in studies by Dr. Dennis Lo, researchers determined that cffDNA was present in maternal plasma in a range of about 3.4%–6.2% (as a percent of total circulating DNA), with the percentages corresponding to early and late pregnancy, respectively. Ex. 2239 ¶¶ 100–101; Ex. 1033, 28–29. Although intact fetal *cells* may also be found in maternal plasma, most fetal DNA in maternal plasma exists in its cell-free form. Ex. 1036, 1612 (noting that, by altering the blood-processing protocol, quantification of total, but not fetal, cell-free DNA is affected).

To analyze cell-free DNA from blood, a blood sample may be first collected (e.g., from a subject's vein) into a tube and then further processed.

See, e.g., Ex. 1003, 10–11 (Plasma Separation Protocol); Ex. 1001, 89:16–20. Then, to separate the cellular components and plasma within the blood sample, a common mode of processing involves the use of a centrifuge. Ex. 1003, 10–11 (disclosing centrifugation, followed by removal of clear plasma above the cell pellet (buffy coat)); Ex. 1033, 16:19–27 (protocol for separating plasma); Ex. 1056 ¶ 25. As explained by Dr. Levy, cell-free DNA may then be further isolated and analyzed with, for example, the use of PCR or fluorescence detection. Ex. 1056 ¶ 22.

Dr. Levy explains that, when processing maternal blood samples, it was routine to use one or more preservative agents in appropriate blood collection tubes. Ex. 1056 ¶¶ 25–26. According to Dr. Levy, one exemplary tube used for this purpose was Becton, Dickinson's Acid Citrate Dextrose ("ACD"), also known as an Acid Citrate Dextrose Solution A ("ACDA"), blood collection tube. *Id.* ¶ 26 (citing references). Such ACD tubes contain multiple compounds, including dextrose—the naturally-occurring form of glucose. *Id.* ("Dextrose is the naturally-occurring dextrorotatory form of glucose, and is often referred to as glucose."); Ex. 1068 ¶ 3.

The '277 patent acknowledges the prior non-invasive use of fetal cells and cell-free fetal DNA, both isolated from maternal blood, for prenatal diagnosis. Ex. 1001, 5:7–59. With regard to fetal cells, the patent notes that the "presence of fetal nucleated cells in maternal blood makes it possible to use these cells for noninvasive prenatal diagnosis," and that such "cells can be sorted and analyzed by a variety of techniques to look for particular DNA sequences." *Id.* at 5:8–13. Yet the patent states that "it is still difficult" to get many fetal cells from maternal blood and "[t]here may not be enough to reliably determine anomalies of the fetal karyotype or assay for other abnormalities." *Id.* at 5:30–34. The '277 patent states that fetal DNA "has been detected and quantitated in maternal plasma and serum" and that "fetal DNA present in the maternal serum and plasma is comparable to the concentration of DNA obtained from fetal cell isolation protocols." *Id.* at 5:39–49. "However," according to the patent, "the diagnostic and clinical applications of circulating fetal DNA is limited to genes that are present in the fetus but not in the mother" and "a need still exists for a non-invasive method that can determine the sequence of fetal DNA and provide definitive diagnosis of chromosomal abnormalities in a fetus." *Id.* at 5:53–59.

The '277 patent describes a method that is said to increase the proportion or percentage of the cffDNA component in a sample from a pregnant female for subsequent analysis. According to the '277 patent, the ability to detect chromosomal abnormalities has been "hindered by the low percentage of free fetal DNA" in maternal samples. *Id.* at 89:1–6. "Increasing the percentage of free fetal DNA would enhance the detection" of genetic abnormalities. *Id.* at 89:6–11.

With the aim of increasing the percentage of cffDNA relative to circulating maternal DNA in a maternal sample, the '277 patent describes adding an agent that inhibits cell lysis. Ex. 1001, 219:38–44 (Example 15) ("[T]he use of cell lysis inhibitors, cell membrane stabilizers, or cross-linking reagents can be used to increase the percentage of fetal DNA in the maternal blood."); *id.* at 89:11–13 ("The percent of fetal DNA in plasma obtained from a pregnant female was determined both in the absence and presence of inhibitors of cell lysis"). The '277 patent explains that, "[w]hile lysis of both maternal and fetal cells is inhibited, the vast majority of cells [in a maternal blood sample] are maternal, and thus by reducing the lysis of maternal cells, there is a relative increase in the percentage of free fetal DNA." *Id.* at 32:36–39. The patent identifies numerous agents as cell lysis inhibitors, cell membrane stabilizers, or cross-linking reagents. *Id.* at 31:57–32:21 (listing, e.g., formaldehyde, formalin, cholesterol, and glucose).

The '277 patent provides results on the use of formalin (formaldehyde in water) as the lysis-inhibiting agent. Ex. 1001, 89:1–91:50 (Example 4), 219:38–226:26 (Example 15).[11] In Example 4, the patent describes collecting a 5 ml blood sample from a pregnant subject, separating the sample into two tubes (each containing EDTA[12]), and adding formaldehyde (25µl/ml) to one of the tubes. *Id.* at 89:18–25. The samples were centrifuged and 800 µl of each maternal plasma sample was then used for DNA purification and further processing to determine the relative amount of cffDNA in the samples. *Id.* at 89:25–91:13. The '277 patent reports that "the percentage of fetal DNA present in the sample that was treated with only EDTA was 1.56%" and the "percentage of fetal DNA present in the sample treated with formalin and EDTA was 25%." *Id.* at 91:14–20. The

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[11] The parties have used formaldehyde and formalin interchangeably in this proceeding, noting simply that formalin is formaldehyde in water. Pet. 2 ("formaldehyde (referred to as formalin when in solution)"), 10 n.4 ("This petition also discusses 'formalin,' which is a liquid that contains formaldehyde and water"); Ex. 2294, 108:4–10 (testimony of Dr. Levy that he is using the terms formalin and formaldehyde "interchangeably").

[12] EDTA is the chemical compound ethylenediaminetetraacetic acid. Ex. 1056 ¶ 56; Ex. 1067, 51:4–5. The '277 patent states that EDTA is a "magnesium chelator." Ex. 1001, 31:52–54.

percent of total cffDNA in eighteen blood samples with and without formalin was then calculated, with the results (mean percentage cffDNA) provided in Table V. *Id.* at 91:35–43 (reporting, *inter alia*, 19.47% with formalin and 7.71% without formalin), 219:38–226:26 (Example 15).

D. Challenged Claims

Independent claims 55 and 81 are illustrative and read as follows:

> 55. A method comprising determining the sequence of a locus of interest on free fetal DNA isolated from a sample obtained from a pregnant female, wherein said sample comprises free fetal DNA and an agent that inhibits lysis of cells, if cells are present, wherein said agent is selected from the group consisting of membrane stabilizer, cross-linker, and cell lysis inhibitor.
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> 81. A method for preparing a sample for analysis comprising isolating free fetal nucleic acid from a the sample, wherein said sample comprises an agent that inhibits lysis of cells, if cells are present, and wherein said agent is selected from the group consisting of membrane stabilizer, cross-linker, and cell lysis inhibitor.

Ex. 1001, 472:66–473:5, 474:52–57.

To illustrate some of the challenged dependent claims, claim 59 depends from claim 55 and recites "wherein said agent is a cell lysis inhibitor," and claim 60 depends from claim 59 and adds that "said cell lysis inhibitor is selected from the group consisting of: glutaraldehyde, derivatives of glutaraldehyde, formaldehyde, derivatives of formaldehyde, and formalin." *Id.* at 473:13–18.

E. Prosecution History

During prosecution, the Examiner rejected pending claims as anticipated by Lo, or for obviousness over Lo combined with other references.[13] Ex. 2064, 1228, 1231. In response, applicant argued that the Examiner had provided no evidence that EDTA in Lo's samples was the claimed "agent" that inhibits cell lysis. *Id.* at 1194–1196 ("[T]he Office has provided absolutely no documentary evidence or rationale in support of its assertion that EDTA is an agent that inhibits cell lysis."), 1199–1200.

Applicant further argued that "the assertion by the Office that EDTA is a cell lysis inhibitor is simply incorrect." *Id.* at 1196. Applicant stated:

> EDTA is not an "agent that inhibits cell lysis." Rather, EDTA is a well-known chelator of calcium and magnesium. EDTA is routinely added to blood during the blood collection process as an anticoagulant due to its ability to chelate calcium. In fact, EDTA is sometimes included as an ingredient in cell lysis buffers. . . . EDTA is clearly referred to as a chelator in Applicant's specification, not as a cell lysis inhibitor (see, e.g., paragraph [0165] of Applicant's specification), whereas multiple examples of agents that inhibit cell lysis are provided separately (see, e.g., paragraphs [0166] to [0167]).

Id. at 1196. As such, applicant argued that Lo's teaching of EDTA does not satisfy the limitation of "an agent that inhibits cell lysis" as claimed. *Id.*

The Examiner withdrew the rejections based on Lo, but entered new rejections for obviousness based on combinations of "Amicucci" or

[13] Claims 55 and 81 correspond, respectively, to pending claims 58 and 87 in prosecution. Ex. 2064, 530. The citations to Exhibits 2064 and 2041 are to the page numbers added to the exhibit copy, not the original pagination.

"Umansky," with "Kiessling." *Id.* at 927–928, 931, 958–961. The Examiner found that Amicucci and Umansky taught all the claim limitations except "an agent that inhibits cell lysis," which the Examiner found was met by Kiessling's disclosure of formaldehyde as a fixative for blood cells. *Id.* at 928, 933, 959, 962.

In response, applicant argued that there was no motivation to combine the newly cited references. *See, e.g.*, *id.* at 574–575. Among other things, applicant argued that the DNA analyzed in the methods of Umansky and Kiessling was "quite distinct" because Umansky analyzed fetal DNA circulating outside a cell, "while the DNA analyzed in Kiessling is in and/or released from a fixed cell." *Id.*; *see also id.* at 593 (advancing similar argument for the Amicucci combination). Applicant also argued that the claimed method addressed a long-felt need and produced unexpected results. *Id.* at 573–574, 591–592. Applicant argued that its method was an alternative to invasive prenatal testing, and that, by adding formalin as an agent that inhibits lysis, the percentage of cffDNA was 25%, compared to 1.56% without formalin. *Id.* at 574 (asserting that, based on prior studies, "the mean percentage of free fetal DNA in a maternal sample was expected to be about 3%").

The Examiner, on September 26, 2007, entered a Notice of Allowability. *Id.* at 523–525. The Examiner stated that the claims were "deemed to be allowable in light of the applicant's amendment filed 30 MAY 07 and the persuasive argument(s) therein." *Id.* at 525.

We also provide an overview of the prosecution of the related '720 patent, which claims similar subject matter involving, at a high level, isolating and detecting free nucleic acid from a sample that "comprises an agent that impedes cell lysis, if cells are present." *See* IPR2021-01271, Paper 1 (reproducing claim 1).

In the '720 patent's prosecution, the Examiner initially asserted Kiessling against the pending claims (but in combination with different references than discussed above for the '277 patent's prosecution). *See, e.g.*, Ex. 2041, 1145, 1149–1157. The Examiner later withdrew those rejections and entered a new round of rejections. Ex. 2041, 2168–2173. The new rejections included combinations based on, *inter alia*, either Amicucci or Holodniy, in further combination with Kiessling, which the Examiner determined taught the addition of formaldehyde as the claimed "agent." *Id.* The Examiner, in making these rejections, stated: "these authors [e.g., Amicucci or Holodniy] do not teach that their samples comprise an agent that impedes cell lysis, if cells are present, and wherein said agent is selected from a defined group which includes a cell lysis inhibitor." Ex. 2041, 2170. Amicucci disclosed that blood samples were collected in EDTA-containing tubes and Holodniy disclosed that blood samples were collected in Vacutainer (i.e., ACD) tubes. Ex. 2046, 1; Ex. 2075, 2.

Applicant further amended the claims, argued that there were no reasons to combine the references, and argued that the claimed methods solved a long-felt need and produced unexpected results. Ex. 2041, 2238, 2249–2250. Ultimately the claims were allowed. *Id.* at 2286–2290 (remarking that the references of record do not alone, or in combination, teach or reasonably suggest the claimed methods).

II. ANALYSIS

A. Principles of Law

"In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable." *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify "with particularity . . . the evidence that supports the grounds for the challenge to each claim")).

To show anticipation under 35 U.S.C. § 102, each and every claim element, arranged as in the claim, must be disclosed in a single piece of prior art. *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359 (Fed. Cir. 2008). "An element may be inherently disclosed only if it is necessarily present, not merely probably or possibly present, in the prior art." *Guangdong Alison Hi-Tech Co. v. Int'l. Trade Comm'n*, 936 F.3d 1353, 1364 (Fed. Cir. 2019). "[A] limitation or the entire invention is inherent and in the public domain if it is the natural result flowing from the explicit disclosure of the prior art." *Schering Corp. v. Geneva Pharm., Inc.*, 339 F.3d 1373, 1379 (Fed. Cir. 2003) (internal quotation and citation omitted).

A claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which that subject matter pertains. *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art;

(2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) secondary considerations of nonobviousness when presented. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966). When evaluating a combination of teachings, we must also "determine whether there was an apparent reason to combine the known elements in the fashion claimed by the patent at issue." *KSR*, 550 U.S. at 418 (citing *In re Kahn*, 441 F.3d 977, 988 (Fed. Cir. 2006)). Whether a combination of elements produces a predictable result weighs in the ultimate determination of obviousness. *Id.* at 416–417.

B. Person of Ordinary Skill in the Art ("POSA")

In determining the level of skill in the art, we consider the problems encountered in the art, the art's solutions to those problems, the rapidity with which innovations are made, the sophistication of the technology, and the educational level of active workers in the field. *Custom Accessories, Inc. v. Jeffrey-Allan Indus., Inc.*, 807 F.2d 955, 962 (Fed. Cir. 1986).

Petitioner proposes the following POSA definition:

> [A POSA] would have had "a M.D. and/or Ph.D. in a related area such as genetics, biochemistry, molecular biology, cell biology, or microbiology and at least one to two years of work in one of those related areas . . . [or] a Bachelor's degree in one of the foregoing areas and at least three to four years of work in that area." . . . Based on this asserted level of skill, a POSA would have had academic or industry experience collecting and analyzing cell-free DNA from blood samples.

Pet. 13 (quoting Ex. 1020 ¶ 15); *see also* Ex. 1056 ¶¶ 15–16.

We applied this definition at the institution stage. Inst. Dec. 39. Patent Owner did not expressly contest this definition during trial, noting only that "Dr. Van Ness meets [this] definition" and "confirms his opinions are the same under either party's definition." PO Resp. 6 (citing Ex. 2239 ¶¶ 20–25). To the extent there are minor differences in the above definition and the alternative definition offered by Dr. Van Ness (Ex. 2239 ¶ 23), we find that Dr. Van Ness's proposal is too broad for at least the reason that it does not require the POSA have experience collecting and analyzing cell-free DNA from blood samples—subject matter relevant to the claims and asserted art. Accordingly, we continue to apply Petitioner's proposed definition here.[14] In any event, the parties' technical declarants meet or exceed either definition and there is no indication that their opinions would change depending on which definition was applied. Ex. 1056 ¶¶ 5–10 (overview of qualifications and experience); Ex. 2239 ¶¶ 4–7 (same).

C. Claim Construction

We interpret a claim "using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. 282(b)." 37 C.F.R. § 42.100(b). Under this standard, we construe the claim "in accordance with the ordinary and customary meaning of such claim as

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[14] We adopted a somewhat differently phrased POSA definition in the Labcorp cases based on the parties' proposals in those matters. *See, e.g.*, IPR2021-00902, Paper 55 at 13–15. On this record, we do not see that such differences are material or would alter the outcome of this Decision.

understood by one of ordinary skill in the art and the prosecution history pertaining to the patent." *Id.*

A patent's specification may provide a special definition for a claim term, or the intrinsic evidence may reveal an inventor's intentional disclaimer or disavowal of claim scope. *Phillips v. AWH Corp.*, 415 F.3d. 1303, 1316 (Fed. Cir. 2005). If the inventor acts as their own lexicographer to give a claim term a special definition, that definition must be set forth in the specification with reasonable clarity, deliberateness, and precision. *Renishaw PLC v. Marposs Societa' per Azioni*, 158 F.3d 1243, 1249 (Fed. Cir. 1998). A disavowal, if any, can be made by language in the specification or the prosecution history. *Poly-America, L.P. v. API Indus., Inc.*, 839 F.3d 1131, 1136 (Fed. Cir. 2016). "To disavow claim scope, the specification must contain 'expressions of manifest exclusion or restriction, representing a clear disavowal of claim scope.'" *Retractable Techs., Inc. v. Becton, Dickinson & Co.*, 653 F.3d 1296, 1306 (Fed. Cir. 2011) (quoting *Epistar Corp. v. Int'l Trade Comm'n*, 566 F.3d 1321, 1335 (Fed. Cir. 2009)). "[T]he standard for disavowal is exacting, requiring clear and unequivocal evidence that the claimed invention includes or does not include a particular feature." *Poly-America*, 839 F.3d at 1136.

Petitioner does not identify any claim terms as needing express construction. Pet. 21–22 ("[T]he Board need not construe any terms of the challenged claims to resolve the underlying controversy."). Patent Owner argues that the claims should be construed to exclude alleged "anticoagulant chelators," including EDTA and ACD, as satisfying the "agent" term. PO Resp. 6–7. We discuss in more detail below.

The two challenged independent claims (claims 55 and 81) both recite, among other limitations, that the sample comprises "an agent that inhibits lysis of cells, if cells are present." *See supra* Section I(D).[15] Patent Owner, citing a claim construction adopted by the district court in Patent Owner's litigation with third-party Natera, argues that this "agent" limitation should be construed the same way by the Board. PO Resp. 6 (citing Ex. 2039, 1). The court interpreted the "agent" limitation to mean:

> [A] substance that inhibits the rupture of cells that is selected from the group consisting of membrane stabilizer, cross-linker and cell lysis inhibitor, and *does not include chelators used as anticoagulants* nor endogenous substances.

Id.; Ex. 2039, 1 (Supplemental Claim Construction Order) (emphasis added).[16] According to Patent Owner, "[t]he specification, prosecution histories [of the '277 and '720 patents], and extrinsic evidence consistently and unambiguously support" this construction. Because "chelators used as anticoagulants" encompasses both EDTA and ACD, Patent Owner argues, the "Board should thus construe the agent term to exclude anticoagulant chelators, including EDTA and ACD." PO Resp. 6–7.

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[15] The related '720 patent includes claims that require "an agent that impedes cell lysis, if cells are present." *See* IPR2021-01026, Ex. 1001, 535:17–18.
[16] Pursuant to our rules, we take this claim construction into account in construing the claims in this proceeding. 37 C.F.R. § 42.100. Although this construction also excludes "endogenous substances," we need not further address that language because no challenge in this case relies on endogenous substances as being the "agent." *Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co. Ltd.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) ("[W]e need only construe terms that are in controversy, and only to the extent necessary to resolve the controversy.") (internal quotation marks omitted).

Patent Owner cites intrinsic evidence in support of its interpretation. *Id.* at 7–14. For example, Patent Owner contends the Specification distinguishes anticoagulant chelators from the claimed agent, including in a working example (Example 4) where a blood sample placed in a tube with *only EDTA* was considered a control and compared with a sample purporting to illustrate the invention, which included *EDTA in combination with formalin*. *Id.* at 7–8 (citing Ex. 1001, 89:11–13, 89:22–23, and 90:62–67). This example, the patent discloses, shows processing of blood samples "both in the absence and presence of cell lysis inhibitors," which Patent Owner contends should be unambiguously read as indicating that an agent that inhibits lysis of cells is "absent" in the EDTA-only sample. *Id.*[17]

Furthermore, Patent Owner cites portions of the prosecution history where the applicant argued that disclosures about EDTA in the applied art did not meet the "agent" term. *Id.* at 9–11 (citing, as discussed *supra* (Section I(E)), applicant's argument that the Office was "simply incorrect" in finding Lo's EDTA was an agent that inhibits lysis of cells; "Rather, EDTA is a well-known chelator of calcium and magnesium . . . routinely added to blood . . . as an anticoagulant due to its ability to chelate calcium."). According to Patent Owner, similar acknowledgments were made during prosecution of the related '720 patent about art that disclosed

[17] As noted above, the '277 patent describes EDTA as a "magnesium chelator" and discusses EDTA separately from the listing of agents that inhibit cell lysis. *Compare* Ex. 1001, 31:52–56 (identifying EDTA), *with id.* at 31:57–32:12 (listing formaldehyde and glucose as, respectively, a cell lysis inhibitor and membrane stabilizer).

processing samples in ACD tubes. *Id.* at 11 (citing Ex. 2041, 1156 (Adams rejection), 2170 and 2254 (Holodniy rejection and response); Ex. 2075, 2 (Holodniy disclosing ACD); Ex. 2023, 20:52–54 (Adams disclosing ACD)). Patent Owner contends that, during reexamination of the ’720 patent, it again disavowed any claim scope that would encompass ACD. PO Resp. 12 n.12 (citing Ex. 2238, 30–34, and *Aylus Networks, Inc. v. Apple Inc.*, 856 F.3d 1353, 1361 (Fed. Cir. 2017)).[18]

Finally, Patent Owner cites extrinsic evidence, including several technical publications as showing that EDTA and ACD are routinely characterized as anticoagulants for blood-collection tubes, not as cell-lysis inhibitors, fixatives, or preservatives. PO Resp. 13–14. Patent Owner cites,

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[18] Although a patentee’s statements to the Office may evidence a disclaimer, Patent Owner’s statements in the cited reexamination do not weigh heavy here. There is no record here the Examiner has acted on Patent Owner’s statements of alleged disavowal in the reexamination. In *Aylus*, by contrast, the district court relied on patentee’s earlier statements made in its preliminary response that contributed to the Board’s *denial of inter partes review* for the relevant claims. *Aylus*, 856 F.3d at 1362–63. In that light, the patentee could not advance a narrow claim interpretation before the Office that secured IPR denial and later maintain a broader (and inconsistent) interpretation before the courts, which the courts in *Aylus* viewed as a prosecution disclaimer. Here, however, the reexamination of the ’720 patent had just begun and we stayed it so the related IPRs could be completed first. We do not treat Patent Owner’s argument *in the present IPR* about the claims not covering ACD or its components as a “disclaimer,” and we decline to elevate similar argument made in the extant yet stayed reexam to such status. *Cf. Cupp Computing AS v. Trend Micro Inc.*, 53 F.4th 1376, (Fed. Cir. 2022) (holding “a disclaimer is not binding on the PTO in the very IPR proceeding in which it is made, just as a disclaimer in a district court proceeding would not bind the district court in that proceeding”).

for example, the FDA labeling for ACD classifying it as an anticoagulant and noting that ACD "acts as an anticoagulant by the action of the citrate ion chelating free ionized calcium." *Id.* at 13 (quoting Ex. 2071, 1). Patent Owner also cites, for example, a published patent application from tube-maker Streck, Inc., that lists EDTA and ACD among the "anticoagulants," while listing preservatives and polysaccharides separately. *Id.* at 14 (citing Ex. 2073 ¶¶ 14 (listing formaldehyde as a preservative), 89 (listing anticoagulants), 90 (listing polysaccharides)).

The question that Patent Owner's proposed claim construction presents is whether a claimed agent that inhibits cell lysis may be a chelator used as an anticoagulant. The Specification and prosecution history make clear that at least one anticoagulant chelator—EDTA—is excluded from satisfying the claimed agent as recited in claims 55 and 81. As discussed above, the Specification describes an example where a sample processed in a tube with only EDTA is characterized as processing a sample in the "absence" of an agent that inhibits cell lysis. Ex. 1001, 89:11–34 (describing, in contrast, the EDTA + formalin sample as showing an example in the "presence of inhibitors of cell lysis"). If EDTA alone could be read as meeting the "agent" limitation, that example would make no sense. Petitioner provides no persuasive argument otherwise.

The prosecution history buttresses the exclusion of the anticoagulant chelator EDTA from being the claimed "agent." For example, as discussed above, the Examiner at one point expressly identified the EDTA in Lo's tubes as satisfying the agent term, prompting applicant's argument that the Office was "incorrect" in doing so, and further distinguishing EDTA as merely "a well-known chelator of calcium and magnesium" that is added to blood "as an anticoagulant due to its ability to chelate calcium." Ex. 2064, 1196 (arguing "EDTA is clearly referred to as a chelator in Applicant's specification, not as a cell lysis inhibitor"); *supra* Section I(E); Ex. 1001, 31:52–56 (listing EDTA and EGTA as chelators). This, in turn, convinced the Examiner to withdraw the finding that EDTA was an "agent" as claimed and move on to art that disclosed formaldehyde as meeting that claim term. *See supra* Section I(E); Ex. 2064, 1024 (Examiner interview summary noting that "applicant pointed out [that] EDTA could not be defined as a cell lysis inhibitor but rather was simply an anticoagulant"). This colloquy during prosecution evidences a clear and unmistakable disavowal of EDTA as being the claimed "agent."

Having determined that Patent Owner disavowed EDTA as the claimed "agent," we next consider whether the disavowal extends more broadly to other anticoagulant chelators and, specifically, to ACD. On this record, we find that it does as explained below. In distinguishing EDTA from being the "agent," Patent Owner relied not on properties that are unique to EDTA, but on the fact that it was a "well-known chelator" used as an anticoagulant in this technical field. Ex. 2064, 1196. Given Patent Owner's clear intent to exclude EDTA from the scope of the claimed "agent" on the basis that it was simply an anticoagulant chelator, we find that a POSA would also have understood other anticoagulant chelators as subject to Patent Owner's disavowal.

Consistent with this understanding, during later prosecution of the related '720 patent, the Examiner found that Holodniy (and at least one other reference) did not disclose the "agent" term—notwithstanding that such references disclosed processing samples in ACD tubes. Ex. 2041, 2170 (finding that Holodniy "teach[es] a method of detecting free nucleic acid which comprises all of the limitations of Claim 1, except these authors do not teach that their samples comprise an agent that impedes cell lysis"); *see also id.* at 1156 (finding Adams "do[es] not teach adding an agent that impedes cell lysis to the sample"). Patent Owner confirmed the Examiner's finding on this point. *Id.* at 2254 (asserting that Holodniy fails to teach or suggest the claimed method "wherein an agent that impedes cell lysis has been added"). And Patent Owner argues persuasively, without dispute from Petitioner, that ACD is the only addition to the collection tubes of Holodniy and Adams besides the sample itself. PO Resp. 11, 16; PO Sur-Reply 2–3; Ex. 2075, 2; Ex. 2023, 20:52–54.[19] This lends further support that neither the Examiner nor Patent Owner considered ACD as an anticoagulant chelator to be an "agent" as claimed.

For the reasons above, we construe the phrase "an agent that inhibits lysis of cells" to exclude chelators used as anticoagulants. Although the "agent" cannot itself be an anticoagulant chelator, the sample, as a whole, in the claimed methods may still *include* chelators used as anticoagulants. The claims use the phrase "said sample *comprises*" indicating that the claims are

[19] It appears to be the case, however, that ACD was never explicitly highlighted by the Examiner or applicant during prosecution in relation to the agent term or otherwise. As we discuss in more detail below, ACD is not a single chemical compound; instead it includes multiple compounds in solution, and some of them are not chelators. Ex. 1068 ¶ 3; Ex. 2071, 1.

open ended, and that the sample may include other unrecited components. *Exergen Corp. v. Wal-Mart Stores, Inc.*, 575 F.3d 1312, 1319 (Fed. Cir. 2009) ("The claim uses the term 'comprising,' which is well understood in patent law to mean 'including but not limited to."). Indeed, as explained above, the Specification includes examples that purport to represent the invention where maternal blood samples are processed in tubes that combine EDTA (an anticoagulant chelator) with formalin/formaldehyde (the cell-lysis inhibitor), which samples are contrasted with a control where the maternal blood is processed in EDTA-only tubes.[20]

D. Overview of the Asserted Prior Art

1. Landes (Exhibit 1003)

Landes is an international patent application that published July 31, 2003. Ex. 1003, code (43). Landes claims priority to a U.S. provisional patent application (No. 60/349,877) that was filed January 18, 2002 (hereafter "the Landes Provisional" (Ex. 1006)).

Landes relates generally to non-invasive methods used to distinguish circulating fetal DNA from maternal DNA in samples from maternal blood, and thereby to detect fetal aneuploidies and alleles. Ex. 1003, code (57).

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[20] Because we find that the intrinsic evidence is sufficiently clear, we need not rely on extrinsic evidence for purposes of construing the claims. That said, based on the present record, consistent with Patent Owner's argument (PO Resp. 13–14), ACD is commonly characterized as an anticoagulant (not a preservative) in many external publications. But that is not always the case, as is plain from other extrinsic evidence of record. *See, e.g.*, Ex. 1062, 613 (describing ACD as a "preservative" for blood).

Landes teaches, for example, that "maternal serum or plasma may be a relatively rich source of fetal DNA based on PCR determinations," that "fetal DNA can be consistently detected in maternal serum as early as 7 weeks," and that such DNA increases during gestation. *Id.* at 2:19–25 (disclosing that "the lowest fetal DNA concentration in plasma as measured by PCR was greater than 20 fetal cell equivalents per mL of maternal blood with some instances where fetal DNA constituted as much as 5% of the total DNA in plasma").

Landes discloses a plasma separation and fetal DNA isolation protocol where maternal blood is first collected into ACDA tubes supplied from Becton, Dickinson. *Id.* at 10:26–12:6. Landes teaches that, following centrifugation and plasma removal, DNA can be isolated from the plasma sample using commercially available kits. *Id.* Landes teaches that the DNA may then be treated with bisulfite to distinguish fetal and maternal DNA and the nucleotide sequence of interest may be subsequently amplified (e.g. via fetal-specific PCR) and analyzed. *Id.* at 12:9–15:3. More specifically, Landes teaches that isolated DNA may be treated "with a reagent which differentially modifies methylated and non-methylated DNA, *e.g.*, bisulfite," which recognizes that fetal DNA is generally "hypomethylated relative to adult DNA." *Id.* at 8:22–28 (explaining that "[o]ne means of generating fetal-specific PCR products is to identify loci that are unmethylated in fetal DNA and methylated in adult/maternal DNA.”).

2. Marx (Exhibit 1029)

Marx is an article that published in 1991. Ex. 1029, 743. Marx relates to techniques that aim to prepare platelet rich plasma samples with reduced white blood cell contamination. *Id.* Marx teaches, for example, “an effective method of significantly decreasing the WBC content” involves “reduced centrifugation time and with the braking programs disengaged.” *Id.*; *see also id.* at 746 (“[N]o brakes gave much more reasonable yields . . . as well as low WBC contamination.”).

3. Lo (Exhibit 1033)

Lo is an international patent application that published in 1998. Ex. 1033, code (43). Lo relates to a detection method performed on maternal serum or plasma from a pregnant female, which comprises detecting the presence of nucleic acids of fetal origin. *Id.* at Abstr. According to Lo, “[t]he plasma or serum-based non-invasive prenatal diagnosis method according to [Lo’s] invention can be applied to screening for Down’s Syndrome” and other chromosomal abnormalities. *Id.* at 5:3–6.

Lo describes methods for non-invasive prenatal determinations of, for example, fetal RhD status from maternal blood samples, which examples involve isolating and analyzing cell-free fetal DNA. *Id.* at 15:4–19:28 (Example 3). Lo discloses that maternal blood samples were collected and centrifuged to separate the plasma from cellular blood components in the buffy coat. *Id.* at 16:20–27. More specifically, Lo teaches collecting blood samples into tubes containing EDTA, centrifuging the samples at 3,000g, carefully removing and transferring the plasma above the buffy coat into plain polypropylene tubes, then centrifuging the plasma samples a second time at 3,000g to again carefully remove and transfer the plasma into fresh, plain polypropylene tubes. *Id.* Lo discloses the samples were then stored at -20 °C until further processing. *Id.* at 15:26–27. Lo's further processing includes, *inter alia*, extracting DNA with the QIAamp kit, using 800 μl of the plasma sample per column, and then performing real time quantitative PCR with the TaqMan system. *Id.* at 17:1–27; *see also id.* at 18–20 (reporting results and discussion).

Lo discloses that the "most important observation in this study is the very high concentration of foetal DNA in maternal plasma and serum." *Id.* at 29:13–14 (reporting that fetal DNA in maternal plasma comprises a mean of 3.4 and 6.2% in early and late pregnancy, respectively). Lo teaches that the absolute concentrations of fetal DNA in maternal plasma and serum are similar. *Id.* at 30:8–9. Lo discloses that the "main difference lies in the presence of a larger quantity of background maternal DNA in serum compared with plasma, possibly due to the liberation of DNA during the clotting process." *Id.* at 30:9–12. According to Lo, while the difference "exerts no noticeable effect on the efficiency of foetal DNA detection using the real time TaqMan system [(i.e., real time quantitative PCR with the TaqMan system)], it is possible that with the use of less sensitive methods, e.g., conventional PCR followed by ethidium stained agarose gel electrophoresis, maternal plasma may be preferable to maternal serum for robust foetal DNA detection." *Id.* at 30:9–17.

4. Valli (Exhibit 1034)

Valli is an article published in 1980 related to studies on various anticoagulants and the effects of such agents on bovine blood in transport. Ex. 1034, 252 ("Various anticoagulants were assessed in order to develop a transport media . . . which would improve leukocyte stability."). Valli reports that "[t]he purpose of this study was to develop a transport medium for bovine blood which would provide stability during routine shipment through the mail without requiring usual precautions against Canadian climatic conditions." *Id.* (explaining that such medium "would allow blood samples for mass screening or for diagnostic purposes to be sent to a central laboratory").

Valli discloses that blood collected in tubes containing EDTA or ACD was tested for leukocyte stability with those agents alone "and in combination with fixative agents (acids, alcohols, and aldehydes)." *Id.*; *see also id.* at n.4 (identifying "formaldehyde and glutaraldehyde"). According to Valli, "pilot studies on various combinations of anticoagulants showed that formaldehyde improved cell preservation but there was marked clumping of platelets and leukocytes after 48 h." *Id.* at 253.

Valli discloses that, "[f]rom the results of this pilot experiment five anticoagulant combinations were selected" for experimentation. *See id.* at 254 (Table I, listing "FORMULATION OF ANTICOAGULANT COMBINATIONS EVALUATED" including Titriplex I and Titriplex IV, both of which are disclosed as including, *inter alia*, EDTA and formalin). According to Valli, "[a]ll three formalin containing preparations (Titriplex I, IV, and ACD-formalin-ASA) were greatly superior to EDTA without additives for maintaining the stability of leukocytes during transport." *Id.* at 257. Valli discloses: "The fact that the results of cell counts . . . were so accurate with formalin containing anticoagulants, indicates that these preparations are deserving of a far wider utilization." *Id.*

E. Ground 1: Anticipation by Landes

Petitioner asserts that claims 55–59, 61–63, 66–69, 80–89, 94, 96, and 126–130 are unpatentable as anticipated by Landes. Pet. 22–42. Petitioner contends Landes is prior art under §§ 102(a) and/or (e). *Id.* at 14–19.[21]

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[21] Before institution, Patent Owner contested the sufficiency of Petitioner's showing on whether Landes was entitled to the priority benefit of the filing date of the Landes provisional application. We disagreed that Petitioner's showing was insufficient. Inst. Dec. 46–51. Patent Owner did not raise any issue during trial about Landes's prior art status and, thus, has forfeited any argument that Landes is not prior art. On this record, we determine that Landes is § 102(a) prior art based on its July 31, 2003, publication date, which precedes the September 11, 2003, filing date of the application that issued as the '277 patent; we do not presume the claims of the '277 patent are entitled to any earlier priority date of other applications listed in its ancestral chain. *PowerOasis, Inc. v. T-Mobile USA, Inc.*, 522 F.3d 1299, 1305 (Fed. Cir. 2008) (explaining that "there is simply no reason to presume that claims in a [CIP] application are entitled to the effective filing date of an earlier filed application"). That is especially true with the '277 patent's complex priority chain, involving several CIPs leading back to provisional applications filed March 1, 2002, and May 8, 2002. Ex. 1001, code (60), (63). Neither party has made any showing in this proceeding that the challenged claims are entitled to such earlier priority dates. Moreover, even if the challenged claims could claim priority to the filing dates in 2002, the Landes provisional (filed Jan. 18, 2002; Ex. 1006) predates those applications and Landes would be § 102(e) prior art based on the reasoning in the Institution Decision, which we adopt here. Inst. Dec. 46–51.

With respect to the "agent that inhibits lysis of cells, if cells are present" limitation of independent claims 55 and 81, Petitioner points to Landes's disclosure of a method for collecting maternal blood in tubes that contain ACD—and, more specifically, that such tubes will necessarily expose the blood to dextrose (i.e., glucose).[22] *Id.* at 25–29 (citing, for example, Ex. 1003, 10:28–29 (disclosing use of an "ACDA blood collection tube"); Ex. 1007, 4:30–39 (listing components of Becton Dickinson's ACD-A formulation, including sodium citrate, citric acid, and 24.5 gm of dextrose•H_2O)). According to Petitioner, "[t]he '277 patent specifically provides that glucose is an agent that inhibits lysis of cells according to the alleged invention." *Id.* at 25–26 (citing Ex. 1001, 32:4–21 ("In another embodiment, an agent that stabilizes cell membranes may be added to the maternal blood samples to ***reduce maternal cell lysis*** including . . . glucose.") (Petitioner's emphasis).

Patent Owner raises multiple counterarguments. PO Resp. 14–34; PO Sur-Reply 4–12. First, essentially repeating its claim construction analysis, Patent Owner argues that Landes uses ACD only as an anticoagulant without disclosing any specific functionality of ACD's dextrose component, and that Landes's ACD should be excluded from satisfying the "agent" term. PO Resp. 15–19; *see also id.* at 24–27 ("[T]he sole role of ACD in Landes was to prevent coagulation so that plasma could be collected and frozen until DNA isolation"). Patent Owner contends that the public has been on notice

[22] The parties did not argue any material distinction between "dextrose" and "glucose" in this proceeding and, like the parties, we use the terms interchangeably herein.

that anticoagulant chelators are not the claimed "agent" and, accordingly, that "Landes's disclosure of ACD (and its dextrose component) fails to meet the 'agent' limitation." *Id.* at 17–19 (citing the aforementioned prosecution history). Second, even if the dextrose of ACD is not excluded from the scope of the "agent" term as a matter of claim interpretation, Patent Owner argues that Petitioner fails to provide evidence sufficient to show that dextrose in Landes's ACD tubes and method necessarily "inhibits lysis of cells" as claimed. *Id.* at 19–20; 26–34 (arguing, *inter alia*, that a "mere possibility" of ACD's dextrose component inhibiting lysis in Landes's method is not enough); PO Sur-Reply 10–12 (arguing that Petitioner's contentions about dextrose regulating "isotonicity" or providing a "nutrient source" for cells are belated and otherwise substantively flawed).[23]

There is no dispute on this record that ACD includes dextrose, and that dextrose is a species of glucose. Ex. 1056 ¶ 26 (Dr. Levy testimony that ACD includes dextrose, the dextrorotatory form of glucose); Ex. 1068 ¶ 3 (Dr. Levy listing components of ACD); Ex. 2074, 4 (FDA prescribing information for ACD A); Ex. 1067 54:2–6 (Dr. Van Ness agreeing that, "[b]y definition, an ACD tube to a person of skill in the art would contain glucose"). Nor is there any dispute that the '277 patent identifies glucose as among several cell-stabilizing agents that may be added to a sample to

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[23] Patent Owner also argues that Landes does not disclose claim 55's step of "determining the sequence of a locus of interest on free fetal DNA isolated from a sample" because Landes's method modifies portions of the cffDNA, changing certain nucleotides (i.e., C to T) in those DNA sequences. PO Resp. 34–36. We do not reach this argument because, as explained herein, Petitioner's challenge to the claims fails for other reasons.

inhibit maternal cell lysis. Ex. 1001, 32:4–21. The dispute here centers on two questions: 1) whether the exclusion of anticoagulant chelators from meeting the claimed "agent" operates to exclude dextrose introduced to a sample through ACD, and 2) whether Petitioner has established, by a preponderance of the evidence, that the dextrose in ACD actually inhibits lysis of cells in Landes's method. We address each of these questions in turn below.

1. Whether exclusion of anticoagulant chelators as the claimed "agent" also excludes glucose introduced to a blood sample in an ACD tube?

As discussed above (Section II(C)), the claims exclude anticoagulant chelators from being the claimed "agent that inhibits lysis of cells, if cells are present." We now consider whether that exclusion extends to the individual components of ACD—dextrose, in particular. First, we provide a brief overview of ACD.

ACD is a solution comprising three main components: (1) citric acid (2) sodium citrate and (3) dextrose. Ex. 1068 ¶ 3 (testimony of Dr. Levy); Ex. 2074 (FDA prescribing information for ACD A), 1 (indicating that the "[g]eneric name" for ACD A is of "citric acid monohydrate, dextrose monohydrate, and trisodium citrate dihydrate," and identifying the "[d]osage form" as "injection, solution"), 4 (describing mechanism of action and identifying the function of each of the three components of ACD). As Dr. Levy explains, "it is the citrate ion chelating free ionized calcium" that acts as an anticoagulant. Ex. 1068 ¶ 3; Ex. 2074, 4 (noting ACD Solution A "acts as an extracorporeal anticoagulant by binding the free calcium in the blood," that "[c]alcium is a necessary co-factor to several steps in the clotting cascade," and that "[s]odium [c]itrate anticoagulates"). The citric acid provides "pH regulation" and dextrose is for "isotonicity"—influencing movement of water across the cellular membrane. Ex. 1068 ¶ 3 (Dr. Levy stating, "citric acid, for pH regulation" and "glucose (*i.e.*, dextrose) for isotonicity"); Ex. 1067, 59:5–20 (Dr. Van Ness testifying to his understanding of "isotonicity"); Ex. 2074, 4 (stating "[c]itric acid for pH regulation" and "[d]extrose for isotonicity"). Dr. Levy testifies that "the glucose of ACD is not the anticoagulant chelator." Ex. 1068 ¶ 3; *see also* Ex. 1067, 56:11–16 (Dr. Van Ness testifying he has no support for glucose of ACD itself acting as an anticoagulant chelator); Ex. 1067, 58:5–15 (Dr. Van Ness testifying he has no basis to disagree with the description of ACD and its mechanism of action in the FDA prescribing information (Ex. 2074)). In short, ACD includes three main components in aqueous solution—a chelator (sodium citrate), a pH buffer (citric acid), and glucose (for isotonicity).

The '277 patent identifies glucose as an agent that inhibits cell lysis. Ex. 1001, 32:4–21 ("[A]n agent that stabilizes cell membranes may be added to the maternal blood samples to ***reduce maternal cell lysis*** including . . . glucose."). This presents the question whether glucose is nonetheless precluded from satisfying the claimed "agent" limitation because it was introduced into the blood sample in a solution that also includes a chelator (sodium citrate) and a pH buffer (citric acid). We find that glucose is not so precluded. As discussed above, the claims are open-ended and may cover combinations that include an anticoagulant chelator (e.g., the compound

EDTA) along with a lysis-inhibiting agent, such as formaldehyde.

The claims place no apparent limitation on how the "agent" comes to be in the sample—no such limitation was argued in this case. The claims simply require that the "sample comprises" an "agent that inhibits lysis of cells, if cells are present." It would seem incongruous that a blood sample to which is added a substance listed as an "agent" in the Specification would be excluded from meeting the claims simply because that substance (glucose) was introduced to the sample as one component of a solution also comprising an anticoagulant chelator (sodium citrate). This is particularly true where there is no indication on this record that glucose (in ACD or otherwise) plays any specific role in anticoagulation or chelation. Ex. 1068 ¶ 3 (Dr. Levy testimony identifying the functions of ACD's components); Ex. 1069, 114:17–20 (Van Ness testifying he would not classify glucose as a chelator).

For Patent Owner's disavowal of claim scope to extend to an individual chemical component of the ACD solution that itself plays no role in anticoagulation or chelation, Patent Owner disavowal needed to make that unambiguously clear. *Poly-America*, 839 F.3d at 1136. As discussed above, sodium citrate of ACD is that solution's anticoagulant chelator—glucose, which provides a different function, is not.[24]

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[24] Patent Owner cites a docket entry in *Ravgen v. Laboratory Corporation of America Holding*, 6:20-cv-00969, as the district court's order granting summary judgment of no anticipation by Landes. PO Sur-Reply 4; Ex. 2306, 1 ("[A]cid citrate dextrose (ACD) in . . . Landes . . . do[es] not disclose the claimed 'agent' . . . as a matter of law."). Beyond noting that

For the reasons discussed above, Patent Owner excluded anticoagulant chelators from being the claimed "agent." *See supra* Section II(C). It is not, however, clear that the exclusion extends to all of the individual components of a multi-part "anticoagulant" solution—at least one of which is specifically identified in the '277 patent as being among the agents that can inhibit cell lysis. As discussed already, during prosecution the Examiner considered at least two prior art references, Holodniy and Adams, each of which collected blood in ACD tubes, and found that those references lacked a teaching of the claimed agent. But there is no indication that in doing so, the Examiner considered the position urged here by Petitioner, i.e., that the glucose sub-component within ACD meets the "agent" limitation. Pet. 25–26 (Petitioner focusing on ACD's glucose, and the listing of glucose in the patent); Pet. Reply 1–2 (distinguishing ACD's citrate component as the anticoagulant chelator separate from glucose and its functionality). We conclude that Patent Owner's disavowal does not unambiguously extend to glucose as relevant to Petitioner's challenge here. Even short of disavowal, we are also not persuaded that the POSA would interpret glucose as excluded from possibly satisfying the "agent" limitation

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the motion was "GRANTED," the order includes one sentence on this point (quoted in relevant part in the parenthetical above) without further explanation of the court's reasoning. Petitioner argues that it was not a party to that litigation, it does not have access to all the briefing and evidence underlying the order, and the court applies a higher standard of proof than is required in this IPR. Pet. Sur-Sur-Reply 1–2. For these reasons, the court's order is of limited persuasive value and plays no role in our analysis.

when other parts of the intrinsic evidence unambiguously list it as a membrane stabilizer and lysis inhibitor.[25]

2. Has Petitioner established that the dextrose component of ACD inhibits lysis of cells in Landes's method

Landes describes processing maternal blood samples in Becton Dickinson's ACDA collection tubes as part of a plasma separation protocol. *See, e.g.*, Ex. 1003, 10:26–32. It is, however, undisputed on this record that there is no express disclosure in Landes about the effects of using those ACD tubes on cell lysis specifically—much less what effect, if any, the dextrose component in those tubes has on cell lysis in Landes's method. Petitioner nevertheless argues that "the method in *Landes* anticipates even though it does not discuss the effect of glucose in the blood sample." Pet. 25–27. According to Petitioner, the '277 patent itself "provides that glucose is an agent that inhibits lysis of cells," and Landes is no less anticipatory regardless of whether Landes intended or recognized that glucose was a lysis-inhibiting agent. *Id.* at 26–29 (citing precedents on inherency); *see, e.g.*, *Schering Corp.*, 339 F.3d at 1377 (holding a "prior art reference may anticipate without disclosing a feature of the claimed

[25] In the remarks where applicant unambiguously disclaimed EDTA, explaining that it was merely a "well-known chelator," applicant then contrasted EDTA with examples of inhibitors of cell lysis expressly listed in the patent, including glucose. Ex. 2064, 1196 (arguing that "EDTA is clearly referred to as a chelator in Applicant's specification, . . . whereas multiple examples of agents that inhibit cell lysis are provided separately (see, e.g., paragraphs [0166] to [0167])"); *id.* at 1381 (paragraph 167, listing "glucose").

invention if that missing characteristic is necessarily present, or inherent, in the single anticipating reference"); Pet. Reply 4 (arguing recognition of an inherent feature is not needed, citing authority).

We agree with Patent Owner that a key issue here is whether "dextrose, when present in ACD under the conditions set forth in Landes, necessarily is the claimed agent." PO Resp. 30. To be sure, the '277 patent states that glucose is a membrane stabilizer that can be added to reduce cell lysis. And that statement is entitled to weight. But, as pointed out by Patent Owner, the patent "never states that ACD (or its components used in combination) disclosed in Landes is a membrane stabilizer that inhibits lysis of cells." *Id.* at 30–31 (arguing that "the mere possibility" of ACD's "dextrose component inhibiting lysis of cells is not sufficient to support inherency"); PO Sur-Reply 6–7 ("Just because the '277 patent identifies glucose as a membrane stabilizer does not mean that glucose under any and all conditions—or specifically in Landes's ACD samples—necessarily inhibits cell lysis.").

In route to deciding this issue, we reject the contention that a mere presence of glucose in Landes's sample is all that must be shown for Petitioner to prevail. As confirmed at oral argument, Petitioner urges an alleged "plain and ordinary meaning" of the claims. Tr. 19:16–20:3. But, in Petitioner's view, it is simply a "capability" of glucose to inhibit lysis of cells under some set of conditions that is needed—no specific level of inhibition, or even *any* inhibition of cell lysis, under Landes's conditions is required. *Id.* at 18:7–12. By the plain language, however, both of method claims 55 and 81 require the sample include an "agent that inhibits lysis of cells, *if cells are present*," and cells are doubtless present in the maternal blood samples taken in Landes's methods. Petitioner cannot carry its burden simply by showing that glucose is introduced to Landes's blood samples via ACD. Rather, Petitioner must show that glucose in Landes's samples actually inhibits at least some cell lysis, although we agree with Petitioner that Landes itself need not have recognized that effect. We also agree with Patent Owner that the claims "do not require just any unspecified 'agent'; they require an agent ***that inhibits lysis of cells***, albeit without requiring that agent to impose a specific amount of inhibition." PO Sur-Reply 7.

Petitioner contends we should accept as sufficient its mere presence-of-glucose position because Patent Owner allegedly charged others with infringement based solely on the addition of an "agent" (formaldehyde or formaldehyde-releasing compounds). Pet. Reply 5 (citing certain testimony of Drs. Chalmers and Van Ness). We decline to do so on this record. Whether some of Patent Owner's witnesses may be applying the claims too broadly here or in other proceedings relative to infringement or objective indicia topics is not decisive on this issue because the intrinsic evidence and plain meaning of the claims controls.[26]

If some cell-lysis inhibition attributable to glucose in Landes's samples is required, Petitioner argues it has, nevertheless, made this

[26] Similarly, the cited 2008 paper by Zhang (Ex. 2009, 1), which published many years after the application that issued as the '277 patent was filed, reporting "no effect" on cffDNA percentages for samples processed at six hours after collection in the presence of formaldehyde cannot change the meaning of the claims. Pet. Reply 6.

showing. Tr. 22:10–22; Pet. Reply 7 ("Ravgen is also incorrect that the glucose in the samples of *Landes* is not an agent that inhibits lysis according to the patent claims."). In support, Petitioner highlights two asserted properties of glucose that are alleged to prevent cell rupture: *isotonicity*, and providing a *nutrient source* to the cells. Pet. Reply 7 (citing, e.g., Ex. 2294, 213:2–214:17 (redirect testimony of Dr. Levy about glucose's properties); Ex. 1056 ¶ 27; Ex. 1068 ¶¶ 2–7; Ex. 2294, 213:2–214:17, 216:18–217:24)); Tr. 14:10–19 (Petitioner's counsel confirming reliance on glucose's two alleged functions: "isotonicity" and "nutrient source").

Patent Owner contends these arguments by Petitioner about glucose's properties are untimely. PO Sur-Reply 10–12. On this record, we do not agree. Isotonicity is discussed in Patent Owner's relied-upon evidence. *See e.g.*, Ex. 2074. We also find that the asserted properties of isotonicity and providing a potential nutrient source are fairly responsive to Patent Owner's argument that cells in the sample could not metabolize glucose under Landes's method, and that Petitioner has otherwise failed to show that glucose inherently inhibits lysis of cells as claimed. It was, thus, appropriate for Petitioner to address these matters further in its Reply. We address the alleged isotonicity and nutrient source properties in turn below.

Starting with isotonicity, Petitioner contends "that glucose 'maintains the isotonicity' of the sample in the presence of the citrate" in ACD. Pet. Reply 7 (quoting Ex. 1067, 60:3–13 (deposition testimony of Dr. Van Ness)); Ex. 1068 ¶ 3 (Dr. Levy noting glucose's isotonicity property). According to Petitioner, "Dr. Van Ness confirmed that, '[i]n the absence of an isotonic solution cells have the potential to rupture.'" Pet. Reply 7 (quoting Ex. 1067, 59:5–20). Petitioner also cites to papers by Loutit (Ex. 1060) and Rapoport (Ex. 1062) as "compar[ing] citrate solution with acid citrate-glucose solution (ACD) and conclud[ing] that the addition of glucose improved cell stabilization." *Id.* (citing Ex. 1068 ¶¶ 4–6); Ex. 1068 ¶ 6 (Dr. Levy opining that "Rapoport explained that citrate creates a hypertonic solution, but with the addition of glucose, the solution 'is actually almost isotonic with blood in terms of effective tonicity, owing to the permeability of the cellular membrane to glucose'") (quoting Ex. 1062, 601–602)).

We agree that the cited evidence on isotonicity supports that glucose in ACD has the ability to inhibit cell rupture by helping to maintain fluid balance within the cells. Rapoport, for example, found that erythrocytes (RBCs) in an acid-citrate, glucose solution underwent osmotic hemolysis more slowly than RBCs in a simple citrate solution. Ex. 1062, 613. Patent Owner cites, however, rebuttal evidence, which we credit, that failure to maintain isotonicity does not *necessarily* lead to lysis. PO Sur-Reply 10–11.

The failure to maintain tonicity causes cells to shrink or swell depending on the solution in which those cells are placed, but does not necessarily result in cell lysis. This is admitted by Dr. Levy on cross examination: "When a cell swells, it doesn't always lead to lysis." Ex. 2299, 31:22–32:12 (explaining it only increases the "potential" for lysis and "it's only when the pressure on the membrane becomes I believe too great that you would get that lysis"); *id.* at 27:4–9 (discussing tonicity); Ex. 1068 ¶ 6 (opining that "without the glucose of ACD, blood cells in a sample would have a greater ***potential*** for cell lysis") (emphasis added).

Dr. Levy also admits other variables can impact whether cells may lyse beyond tonicity. Ex. 2299, 33:19–25. Yet Dr. Levy did not consider such variables or conditions to show persuasively how the glucose of ACD under Landes's conditions, would necessarily inhibit cell lysis in Landes's method where samples are promptly processed within minutes and frozen. *Id.* at 37:19–22 (did not consider pH), 40:4–9 and 41:6–42:10 (did not consider temperatures); PO Resp. 32–34 (noting differences in, *inter alia*, processing times and temperatures, and glucose concentration compared to Landes); PO Sur-Reply 8–11[27] (arguing the conditions of Petitioner's references do not resemble Landes's conditions); Ex. 1060, 3 ("Blood was stored . . . for 4 weeks" in Loutit); Ex. 1062, 594 (storage of whole blood samples in a refrigerator at 4° C, followed by transport to a laboratory for testing over the course of days and weeks, in Rapoport); Ex. 1003, 10:28–11:2, 16:20–24, 19:19–23 (Landes's conditions where plasma is separated from the cellular fraction within about 20 minutes and promptly frozen at - 80 °C).

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[27] Patent Owner suggests Dr. Levy did not consider glucose concentrations (PO Sur-Reply 11 (citing Ex. 2299, 39:11–21)), but that is not entirely accurate. Dr. Levy opines that the glucose concentration of ACD in Landes is "comparable" to the glucose concentrations in some cited references. Pet. Reply 7–8; Ex. 1068 ¶ 7 (comparing 24.5 g/L of the Becton Dickinson ACDA tubes to the concentrations of 27.3 g/L and 30 g/L of Loutit and Rapoport (Ex. 1060, 186 and Ex. 1062, 593, respectively); *but see* Ex. 2299, 69:14–25 (Dr. Levy admitting, when asked whether he compared the concentrations of components in Rapoport's solution to concentrations of ACDA (as in Landes) that he "didn't look to draw an opinion on that").

Turning next to glucose as a "nutrient source," this is not taught in Landes and Petitioner has not shown that glucose necessarily provided a lysis-inhibiting nutrient source in Landes's method. Petitioner's evidence, including Mollison,[28] provides some support that glucose can be a nutrient source leading to increased RBC survival, and decreased cell breakdown and lysis, but we agree with Patent Owner and Dr. Van Ness that Mollison is discussing those effects upon longer-term storage of blood samples that are not applicable to Landes.[29] Ex. 1011, 16 (disclosing that glucose "can also support red cell metabolism during storage"); Ex. 1067, 75:24–76:5 (Van Ness testimony that "glucose can be a nutrient for cells"); PO Resp. 33 (citing Mollison's teaching that samples were stored for weeks); Ex. 2239 ¶ 137 (Van Ness's testimony that "Mollison's disclosure is not applicable to . . . Landes, where cells are not stored . . . over a four week span").

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[28] Mollison, P.L., *Historical Review: The Introduction of Citrate as an Anticoagulant for Transfusion and of Glucose as a Red Cell Preservative*, British J. of Heamatology, 108:13–19 (2000) (Ex. 1011 or "Mollison"). Mollison provides a historical review of the development of citrate and citrate glucose compositions arising in connection with blood transfusions and experiences during World War I and afterward. *See generally* Ex. 1011 (discussing citrate as an anticoagulant and glucose as a preservative: "[T]he use of citrate made it possible to separate donor and recipient in space; the use of glucose made it possible to separate donation and transfusion by a substantial period of time.").

[29] Petitioner cites several references as purporting to show glucose's "known" properties. Pet. 16 (citing, e.g., Ex. 1011 (Mollison), Ex. 1019 (Lee), Ex. 1022 (Carter)). But Petitioner's citation is conclusory, providing no substantive, supporting explanation pertinent to the present challenge. Dr. Levy's corresponding testimony on this point cited by Petitioner is equally vague and unhelpful. Ex. 1056 ¶ 27.

Relevant to Petitioner's "nutrient source" theory, another condition, besides time, is temperature. On cross-examination, Dr. Levy testified that "it's possible" cells "would be able to metabolize glucose [even] when they were stored at 3 to 7 degrees C." Ex. 2299, 62:15–25. Dr. Levy further testified that "the colder you get, the slower biochemical reactions occur" (*id.*) but ultimately conceded that "I didn't look at these things or was asked to consider these things for . . . refuting or discussing the issues that I disagreed with Dr. Van Ness" (*id.* at 63:4–7); *see also id.* at 64:2–16 (admitting "I wasn't asked to consider temperatures" on the question of whether cells are nonfunctional at low storage temperatures). Petitioner provides no persuasive evidence to indicate that any cells in Landes's samples were inhibited from lysing by virtue of cells metabolizing glucose under Landes's conditions. Quite the opposite, Dr. Levy admitted he could not say if any cells in Landes's method used glucose as a nutrient and avoided lysis as a result. *Id.* at 59:25–60:7 (Q. "Do you know if the cells would have used that glucose and metabolized it as a nutrient source in the diluents 1 to 8 [of Landes]? A. That wasn't the focus of what I looked into, and that wasn't directly measured in this article.").

Patent Owner, on the other hand, cites evidence that cells are incapable of using glucose as a nutrient source under certain conditions—especially at low temperatures. PO Resp. 27–28. Noting McMillan (Ex. 2235, 662–665) and its disclosure that "[r]apid separation of the sample or cooling will also prevent glycolysis [(i.e., cellular metabolism of glucose)]," Dr. Van Ness opines that the POSA would understand that "cells collected by Landes did not—and could not—metabolize or use the dextrose component of ACD as a membrane stabilizer or cell lysis inhibitor." Ex. 2239 ¶¶ 132–133. That is so, Dr. Van Ness explains, because Landes's method rapidly processes the blood to separate plasma after which the plasma is transferred to a fresh tube and promptly frozen at -80°C. *Id.* ¶¶ 130–131. And Dr. Levy agrees that, from Landes's disclosure, this process takes only "twenty plus a couple of minutes." *Id.* (citing Ex. 2294, 151:19–152:7). Petitioner dismisses McMillan because it analyzed the cells' inability to metabolize glucose already present in the blood, not glucose separately added as part of ACD. Pet. Reply 8. But if the cells are rendered nonfunctional and unable to consume endogenous glucose as a nutrient source at low temperatures, as suggested by McMillan and Dr. Van Ness, Petitioner fails to provide evidence or persuasive technical reasoning to suggest the cells' behavior would change if more glucose were added. Ex. 2239 ¶¶ 132–133, 137; Ex. 1068 ¶ 8.

Altogether, considering the argument and evidence of record, Petitioner has shown, at best, that it is *possible* for glucose to help inhibit cell lysis by reason of isotonicity and providing a nutrient source. But such possibilities are not enough for Petitioner to prevail on its challenge to the claims as anticipated by Landes. *In re Oelrich*, 666 F.2d 578, 581 (CCPA 1981) ("Inherency, however, may not be established by probabilities or possibilities. The mere fact that a certain thing *may* result from a given set of circumstances is not sufficient.") (internal quotation and citation omitted); *Guangdong Alison Hi-Tech*, 936 F.3d at 1364 (same). Petitioner has not, on this record, met its burden to show that the glucose component of ACD in

Landes's specific method is an "agent that inhibits lysis of cells" in the manner claimed.

Lastly, we address the argument about the comparative effects of EDTA and ACD. In its Response, Patent Owner cites a handful of references reporting no significant differences between use of EDTA and ACD as anticoagulants to prepare plasma samples as support for the proposition that the dextrose of ACD has not been shown to inhibit lysis of cells. PO Resp. 26–27 (citing, e.g., Ex. 1019, 5 (finding "no significant difference between EDTA and ACD plasma")) (emphasis omitted); Ex. 2237, 2 (reporting "no statistically significant difference between $\log_{10}$ copy numbers in samples collected in ACD tubes and those samples collected in EDTA tubes")). Petitioner responds that this is an improper attempt by Patent Owner to read a "significant" lysis-inhibiting effect into the claims. Pet. Reply 5–6 (asserting such reading is "indefinite"). We disagree that was Patent Owner's purpose. Instead, we understand that Patent Owner sought to show that EDTA (without dextrose) and ACD (with a dextrose element) were materially indistinguishable in terms of the reported results as anticlotting agents. PO Resp. 27.

This argument and evidence about comparing results with EDTA and ACD is ultimately neutral to our analysis. We have not been directed to evidence that calls out any specific effect of dextrose in these references. As discussed above, EDTA is a chelator that prevents coagulation of the blood. ACD is also an anticoagulant, but includes a different chelator compound (sodium citrate/citrate ion). The argument and comparisons here between EDTA and ACD, which provide substantially the same anticoagulant effects using different chelators, sheds little light on whether and to what extent any lysis inhibition is attributable to glucose in ACD, much less whether glucose had such an effect in Landes's method. Put differently, this argument and evidence fails to show specific effects of dextrose in Landes's ACD samples separate and apart from ACD's anticoagulation function generally.

3. Conclusion on Ground 1

For the reasons discussed above, we find Petitioner has not established by a preponderance of the evidence that Landes discloses "an agent that inhibits cell lysis, if cells are present" as recited in independent claims 55 and 81, and the claims depending therefrom. Accordingly, Petitioner has not established that claims 55–59, 61–63, 66–69, 80–89, 94, 96, and 126–130 are anticipated by Landes.

F. Ground 2: Obviousness over Landes and Marx

Petitioner asserts that claim 95 would have been obvious over Landes and Marx. Pet. 42–46. The art is summarized above. Claim 95 depends from claims 94 (requiring "isolation of nucleic acid comprises a centrifugation step") and independent claim 81 (discussed above under Ground 1), and claim 95 adds that "the centrifugation step is performed with the centrifuge braking power set to zero." Ex. 1001, 475:30–32. Petitioner asserts that Marx discloses zero-brake centrifugation as a technique that would have been obvious to include in Landes's method to better avoid disturbing the buffy coat when processing plasma samples. Pet. 43–45. Patent Owner's arguments against Ground 2 are the same as addressed above on Ground 1. PO Resp. 14 (arguing "Grounds 1–2" together).

Although not explicitly stated, it is apparent that Petitioner relies on the dextrose in Landes's ACD for satisfying the "agent" limitation of independent claim 81. Pet. 42–46 (referring to Marx's teachings about centrifugation speeds/braking without identifying any disclosure in Marx as allegedly satisfying the "agent" or other limitations of claim 81). For the reasons explained above under Ground 1, Petitioner has not established by a preponderance of the evidence that Landes discloses claim 81's "agent" limitation. As Ground 2 relies on Petitioner prevailing on the threshold showing for claim 81 under Ground 1, Ground 2 also fails here.[30]

G. Grounds 3 & 4: Obviousness over Lo and Valli; Lo, Valli, and Marx

Petitioner asserts, for Ground 3, that claims 55–63, 66–69, 80–91, 94, 96, 126–130, 132, and 133 would have been obvious over Lo and Valli. Pet. 46–65. Petitioner asserts, for Ground 4, that claim 95 would have been obvious over Lo, Valli, and Marx. Pet. 65–66. Lo, Valli, and Marx are summarized above. As discussed below, Grounds 3 and 4 turn, in this case, on whether Petitioner met its burden to demonstrate, by a preponderance of the evidence, that a POSA would have been motivated and found it obvious to modify the asserted prior art in the manner proposed by Petitioner.

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[30] We need not reach Patent Owner's argument on objective indicia of nonobviousness. *See Hamilton Beach Brands, Inc. v. f'real Foods, LLC*, 908 F.3d 1328, 1343 (Fed. Cir. 2018) (holding "objective indicia of nonobviousness" "need not [be] address[ed]" because the court "affirm[ed] [the] Board's findings regarding the failure of the prior art to teach or suggest all [claim] limitations").

We start with Petitioner's contentions, citing those against claim 55 as representative. We then outline Patent Owner's counterarguments before moving to our analysis, which applies equally to claims 55 and 81.

1. Petitioner's Contentions

Petitioner argues that Lo discloses all claim 55's limitations except for the "agent that inhibits lysis of cells" limitation, which Petitioner asserts is disclosed in Valli's teaching of formalin/formaldehyde. Pet. 46–48.[31] According to Petitioner, Lo describes a method for isolating and analyzing cell-free DNA from a pregnant female's blood, and determining the sequence of a locus of interest on isolated cell-free fetal DNA as claimed. *Id.* at 46–47. In particular, Petitioner identifies Lo's methods "for isolating cell-free fetal DNA from a sample of maternal blood in Examples 3 and 5." *Id.* at 46 (citing Ex. 1033, 15:4–20:3, 23:14–33:30). Petitioner cites Lo's plasma "sample preparation" protocol where maternal blood is collected in EDTA-containing tubes, centrifuged, DNA is extracted with the QIAamp blood kit, and "Real time quantitative PCR" is used to determine fetal rhesus D ("RhD") status or for quantitative analysis of cffDNA in plasma. *Id.* at 46–47 (asserting the real-time PCR step "determines the sequence of a locus of interest") (citing, e.g., Ex. 1033, 16:10–17, 17:1–6, 10:23–12:29, 25:19–27; Ex. 1056 ¶¶ 105–106).

[31] Petitioner qualified this lack of disclosure of the "agent" limitation in Lo's cited method by stating "[t]o the extent EDTA is not" the agent. Pet. 47–48. We do not, however, understand Petitioner's position in this case to be based on EDTA meeting the agent limitation and, as explained above (Section II(C)), applicant clearly disclaimed EDTA as the "agent."

According to Petitioner, a POSA would have understood from Lo, Valli, and other references that WBCs in a blood sample may become unstable and subject to lysis. Pet. 48. Petitioner contends "[a] POSA would have understood that unstable maternal leukocytes (or white blood cells) in a blood sample could release maternal cell-free DNA, which would make 'robust foetal DNA detection' more difficult." *Id.* (citing Ex. 1033, 30:8–17; Ex. 1019, 276, 279; Ex. 1056 ¶¶ 108–109). Petitioner contends a "POSA would have had motivation to combine *Lo* and *Valli*" and "would have recognized the issue of maternal cell lysis in a blood sample." *Id.*; Ex. 1033, 30:8–17 (disclosing a larger quantity of background maternal DNA in serum compared to plasma, "possibly due to liberation of DNA during the clotting process"); Ex. 1035 (Hahn), 1577 ("The contamination of the plasma sample by maternal cells or cell remnants would, however, affect the apparent concentration of maternal DNA in the sample."); Ex. 1036 (Chiu), 1612 (noting that blood processing protocols may affect total DNA concentrations but not the amount of fetal DNA); Exs. 1038, 196 and 1039, 33 (discussing apoptosis of cells during pregnancy); Ex. 1056 ¶ 109.

Petitioner contends that Valli and Lo both relate to preparing and screening blood samples, and that Valli assessed blood collection agents for their capacity to stabilize bovine leukocytes (WBCs) during transportation of the samples. Pet. 48–50 (citing Ex. 1034, 252). Petitioner contends Valli "exemplifies the decades of experimentation" with blood stabilization. *Id.* at 49 (citing Ex. 1034, 252–253 and other references as allegedly related to use of formaldehyde as a fixative). According to Petitioner, Valli "tested two formalin-EDTA preparations (Titriplex I and IV) and ACD-formalin-ASA."

Id. at 51 (citing Ex. 1034, 253, 257, Table 1). Valli allegedly "concludes that formalin-EDTA preparations tested [were] 'greatly superior to EDTA without additives for maintaining the stability of leukocytes during transport.'" *Id.* (citing and quoting Ex. 1034, 257 (disclosing that "these preparations are deserving of a far wider utilization"); Ex. 1056 ¶ 114).

Thus, Petitioner contends, "[a] POSA following *Lo* to isolate cell-free fetal DNA from a maternal plasma in the presence of EDTA . . . [(citing Lo's Examples 3 and 5)] would have been motivated to substitute a formalin-EDTA preparation, as suggested by *Valli*, to improve white-blood-cell stability (thereby reducing background cell-free maternal DNA resulting from cell lysis)." Pet. 52. And, Petitioner argues, "based on the results described in *Valli*, the POSA would have applied the prior art methods of using formalin to the method of isolating cell-free DNA in Lo to yield predictable results with a reasonable expectation of success." *Id.* (citing, e.g., Ex. 1056 ¶¶ 107–115 (Dr. Levy testifying that a POSA "would have been motivated to try to substitute EDTA with a formalin-EDTA preparation" and that "using formalin applied to *Lo's* method of isolation of cell-free fetal DNA, would have produced predictable results, where the [POSA] would have reasonably expected success.")).

2. <u>Patent Owner's Counterarguments</u>

Patent Owner raises several counterarguments. First, Patent Owner argues that a POSA would not have been motivated to modify Lo to address alleged concerns with cell lysis in the manner urged by Petitioner. PO Resp. 36–38. Patent Owner contends there is nothing in Lo that would suggest cell lysis (or resulting contamination) is a problem in Lo's methods, especially when plasma is used. *Id.* To the contrary, Patent Owner argues, Lo's methods (including those that use EDTA as an anticlotting agent) provide "*robust* foetal DNA detection," as admitted by Petitioner. *Id.* (citing Pet. 8, 12, 20, 48); Ex 1033, 30:12–24; Ex. 2239 ¶¶ 149–150 (testifying that Lo, even if contamination had been thought a possible problem, teaches the use of plasma instead of serum would be sufficient to address it; also testifying that Lo's methods allowed for robust detection whether serum or plasma was used); PO Sur-Reply 13–14 (citing Ex. 1033, 30:8–20).

Further to this argument, Patent Owner asserts that Lo's approach for preparing plasma was accepted in the field as allowing detection of the percentage of free fetal DNA present in circulating (*in vivo*) maternal blood. PO Resp. 37 (citing Ex. 1033, 28:13–17, 29:21–23; Ex. 1035, 1); Ex. 2239 ¶¶ 150 (testifying about Lo's disclosure of the mean free fetal DNA fraction of 3.4% and 6.2% in early and late pregnancy, respectively), 153–154 (testifying that "[i]t was widely accepted at the time . . . that fetal DNA concentration in maternal plasma ranged from 3.4% to 6.2% . . ., as previously reported by Lo"). According to Patent Owner, it was not until several years later—after the present invention—that researchers in the field, including Drs. Lo and Chiu, realized that even greater proportions of cffDNA were available for detection. PO Resp. 37–38 (citing, e.g., Ex. 2114 (paper by Chiu and Lo from 2008), 4 ("approximately 2-fold higher than previously reported"); Ex. 2121 (paper by Chiu and Lo from 2011), 6 (reporting "15.2%"); Ex. 2124, 2–3 ("two- to threefold higher"). As Lo's data on the available cffDNA percentages (mean ~3–6%) was widely reported and accepted, including by the key researchers in the field, Patent Owner contends this undermines Petitioner's position that the POSA would have obviously modified Lo's method in an attempt to further reduce total background DNA and increase the free fetal fraction in the sample. PO Resp. 37–38; PO Sur-Reply 13; Ex. 2239 ¶¶ 154–155 (testifying that, based on the state of the art, existing approaches, like in Lo, already provided for maximal recoveries, as then understood).

Although researchers in the field like Lo, Chiu, or Hahn, may have continued to refine their techniques, Patent Owner contends that, prior to the invention of the '277 patent, none taught or suggested the solution Petitioner argues would have been obvious. PO Resp. 37–38; PO Sur-Reply 13–14. Instead, Patent Owner contends, Chiu and Hahn, suggest only physical processing techniques (centrifugation and/or filtration) to clear cells or cell remnants from the sample. PO Resp. 37 (citing, e.g., Ex. 1036, 1607–1608, 1613 (disclosing centrifugation/microcentrifugation or centrifugation with filtration can produce "[v]irtually cell-free plasma")). Patent Owner contends that none of these prior art references suggests any reason to inhibit cell lysis before or during plasma separation, or even after separation during additional plasma processing. *Id.* (citing Ex. 2239 ¶ 153). And, Patent Owner argues, none "hint that *compositions in the tube* could or should be adjusted" to improve the initial plasma processing steps—let alone to inhibit cell lysis. *Id.* at 37–38 (citing Ex. 1036, 2; Ex. 2239 ¶ 153).

Second, Patent Owner argues that a POSA would not have turned to Valli for modifying Lo's method as proposed. PO Resp. 39–42. Patent Owner contends Valli does not discuss DNA, cell-free DNA, or even cell lysis. *Id.* at 39 (arguing "cell counts are not the same as cell lysis measurements" and noting Valli's "inconclusive" data, some reporting that cell numbers increased; citing Ex. 1034, Tables II–III; Ex. 2239 ¶ 158). Patent Owner contends Valli's pilot experiment, which forms a basis for Valli's conclusions, is refuted by non-pilot studies cited in Valli itself. *Id.* at 40 (explaining non-pilot studies found "total white cell counts did not decrease in EDTA-treated blood stored at 4°C"—opposite to Valli's pilot; citing Ex. 2102 (DeLange and Lampasso), Fig. 1); *see also id.* (asserting Valli's conclusions are flawed); Ex. 2239 ¶ 159 (testifying Valli's confidence intervals are so large some results are not statistically distinguishable). Patent Owner argues Valli's conclusions about EDTA + formalin blood anticoagulants versus EDTA alone are further called into question because Valli reports testing of Titriplex I and Titriplex IV, both of which *include* other anticoagulant compounds and *exclude* EDTA and formalin. PO Resp. 40–41 (citing, e.g., Ex. 2097, 32 (disclosing that Titriplex I is nitrilotriacetic acid (NTA)) and Titriplex IV is DCTA)). Lastly, Patent Owner argues a POSA would have been skeptical of Valli in view of its inconsistent and flawed data and regarded its disclosures about stabilizing cow's blood for shipment under Canadian climate conditions as inapplicable to Lo's methods for analyzing cfDNA. *Id.* at 41–42 (asserting that "ordinary artisans never adopted Valli's 1980 approach for cow blood transport").[32]

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[32] Patent Owner argues the POSA would not reasonably have expected success in combining Valli with Lo for essentially the same reasons. PO Resp. 42–43.

Third, Patent Owner argues the POSA would not have been motivated to use formaldehyde in Lo's method, or expected success in doing so. PO Resp. 44–47. Patent Owner notes that "[e]vidence suggesting reasons to combine cannot be viewed in a vacuum apart from evidence suggesting reasons not to combine" even if such evidence is found to not "rise to the level of teaching away." *Id.* at 44 (quoting *Arctic Cat Inc. v. Bombardier Recreational Prods. Inc.*, 876 F.3d 1350, 1363 (Fed. Cir. 2017)). Patent Owner then cites several technical sources reporting on formaldehyde's dangers, including its known capacity to damage DNA. *Id.* at 44–47.

Patent Owner contends "sources showed concerns regarding formaldehyde and its potential to detrimentally affect nucleic acids" that would have discouraged the POSA from adding it to Lo's methods. *Id.*; Ex. 2239 ¶ 167. For example, Patent Owner contends that "major players" in the blood collection tube market, such as Streck, warned that formaldehyde was "*toxic*, *flammable*, *and carcinogenic*," in addition to other problems. PO Resp. 44–45 (citing Ex. 2230 (Smith patent), 2:37–43 (describing formaldehyde as expensive, inconvenient, and dangerous to lab workers)). Even where formaldehyde was used, Patent Owner contends that "it was as a fixative for applications studying *cells*, not cell-free DNA." *Id.* at 45 (citing Ex. 2239 ¶¶ 169–173 (testifying, *inter alia*, that before Lo, the non-cellular plasma fraction was "routinely discarded" by investigators studying prenatal diagnoses (quoting Ex. 1033, 2:5–9))). Patent Owner also cites studies reporting on formaldehyde's "detrimental mechanisms" and "effects on DNA" even in the context of cells or cellular DNA. *Id.* (citing Ex. 2139 (Swenberg), 1). Indeed, Patent Owner cites evidence allegedly warning against formaldehyde's use and recommending other fixatives altogether. *Id.*; Ex. 2150 (Srinivasan), 6 ("A method to overcome the problems of formaldehyde is to use an alternative fixative that is better suited for the preservation of nucleic acids and proteins.").[33] Accordingly, Patent Owner contends that "significant evidence of negative perceptions of formaldehyde" would have dissuaded the modification of Lo proposed by Petitioner. PO Resp. 46–47; PO Sur-Reply 18–20.[34]

3. Analysis

On the complete record developed through trial, we agree with Patent Owner and find that Petitioner has not carried its burden to establish that a POSA would have been motivated to modify Lo's plasma processing and cffDNA detection method to include Valli's formalin/formaldehyde in the manner proposed by Petitioner. We explain below.

We find that Petitioner's proffered motivation to change Lo's methods to remedy the asserted problem of cell lysis is, at best, weak. There is no dispute, on this record, that potential for lysis of cells in a maternal blood

[33] Patent Owner cites several references as evidencing the problems with formaldehyde's use, including damage to nucleic acids, but some of the references post-date the filing date of the '277 patent by years. *See, e.g.*, PO Resp. 46–47 (citing Ex. 2155, published in 2005); Ex. 2047, published in 2018. For purposes of this decision, we do not rely on these later references.
[34] Patent Owner also argues that Petitioner's analysis for several dependent claims is infected with hindsight bias (PO Resp. 47–48) and that objective evidence of non-obviousness (e.g., unexpected results) support the patentability of the challenged claims (*id.* at 48–64). We need not reach these arguments because Petitioner's challenge to the claims fails for other reasons discussed herein.

sample was a known issue. Lo itself comments on larger quantities of maternal DNA released in serum compared to plasma. Ex. 1033, 30:10–12. As for the cause, Lo teaches the larger DNA content in serum may be "due to the liberation of DNA during the clotting process." *Id.* This was similarly recognized by others in the field. *See., e.g.*, Ex. 1035, 1577 ("[T]he amount of maternal circulatory DNA was significantly increased in serum samples compared with plasma because of the liberation of maternal DNA from maternal cells lysed during clotting."); Ex. 1019, 279 ("The most likely explanation for the higher levels of genomic DNA in serum than in plasma is that the process of clotting lyses WBCs, which release nuclear fragments into the serum."). Petitioner, however, relies on Lo's *plasma* separation protocols, which involve the collection and prompt processing of samples in blood collection tubes that include the anticoagulant EDTA, mitigating concerns with clotting and lysis resulting from it. Pet. 46–47; Ex. 2239 ¶¶ 149–150 (testimony of Dr. Van Ness that "Lo teaches that using plasma instead of serum would be sufficient to avoid contamination by cell lysis"); Ex. 1067, 24:24–25:12 (testimony of Dr. Van Ness that, while the POSA would have understood that lysis was an issue generally, and understood its potential to decrease cffDNA percentages, the POSA would also have known that Lo addressed such lysis by adding an anticoagulant).

Continuing, Lo suggests it is possible cell lysis might be a concern if "less sensitive" DNA detection methods were used. Ex. 1033, 30:14–17. But, again, Lo teaches that such concerns can be addressed by using plasma instead of serum. *Id.* (disclosing that "maternal plasma may be preferable to maternal serum" for fetal detection if a "less sensitive" method, like "conventional PCR" were used). Moreover, Petitioner's challenge does not urge that Lo's method be changed to use less sensitive cffDNA detection techniques. To the contrary, Petitioner relies on Lo's "[r]eal time quantitative PCR" (*see* Pet. 47), which Lo teaches provided for robust detection of cffDNA—whether in plasma or serum. Ex. 1033, 10:23–11:13 (describing use of real time quantitative PCR using the TaqMan system); Ex. 2239 ¶ 150 (Dr. Van Ness testifying persuasively that Lo's methods allowed robust detection of cffDNA in either plasma or serum). Indeed, Lo suggests strongly that lysis and any associated increased background DNA would not be a problem with its highly-sensitive methods because, even with the additional maternal DNA observed in serum samples, "this [larger quantity of background maternal DNA] exerts *no noticeable effect* on the efficiency of foetal DNA detection using the real time TaqMan system." Ex. 1033, 30:9–17 (emphasis added).

There is other argument and evidence of record that the blood processing methods chosen may impact total DNA concentrations, namely concerning the disclosures of Chiu and Hahn. Exs. 1035, 1577; Ex. 1036, 1612. This evidence suggests that some residual cells might remain and be subject to lysis, even in plasma. Hahn, for example, teaches "care must be taken" in clearing the plasma of cells and cell remnants. Ex. 1035, 1577–1578. But Lo already took care in processing its samples to separate the plasma and cells. Ex. 1033, 16:20–27 (disclosing multiple centrifugation steps at 3,000g where, at each step, plasma was "carefully removed" before being transferred to fresh polypropylene tubes and frozen at -20°C for subsequent processing; "Great care was taken to ensure that the buffy coat was not disturbed" in this process). We also agree with Patent Owner and Dr. Van Ness that neither Chiu nor Hahn persuasively support modifying the blood collection tube's contents to add further chemical agents because both teach that physical processing addressed cell contamination. PO Resp. 37; Ex. 2239 ¶ 151 (Dr. Van Ness testifying that art at the time taught "physical processing" could fully address issues concerning maternal cell contamination).[35]

Petitioner responds, repeating that the "problem" of cell lysis was known, and further contends that the law normally attributes to the POSA a desire to improve upon what was known. Pet. Reply 10–11 (citing Ex. 1056 ¶¶ 24, 47–48, 109–110). We do not disagree. Petitioner's argument, however, begs the question *why* the POSA would have believed that lysis of maternal cells was a problem, not in a more general sense (as noted in the testimony of Dr. Levy cited in support), but a problem that remained with the specific methodology of Lo relied upon by Petitioner in its challenge—including the use of carefully processed plasma samples and real time

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[35] Chiu indicates that a single slower-speed centrifugation step at 800g or 1,600g resulted in plasma with higher total DNA concentrations, but if that initial centrifugation step at 1,600g was followed by another centrifugation at 16,000g (microcentrifugation) or a filtration step, that "[v]irtually cell-free plasma" could be obtained. Ex. 1036, 1608, 1612–13. This, however, casts little light on whether a POSA would have been concerned with residual maternal cells and lysis in Lo's method, which used multiple centrifugation steps at 3,000g as part of its careful plasma processing as discussed. Chiu also does not appear to compare its results or methods with Lo's, and Petitioner provides no analysis of the differing methods, to show persuasively why the POSA would still think that cell contamination and lysis would have remained a concern in Lo.

quantitative PCR detection. Petitioner does not provide persuasive argument or evidence that answers that question. As to the assertion (*id.* at 11) that, at the time of the invention, the percentage of cffDNA that was then believed to be available was "too small to render accurate detection or diagnosis," that assertion is at odds with at least Lo, which disclosed robust detection and screening for several conditions including Down's Syndrome. *See* Ex. 1033, 5:3–6. Moreover, Patent Owner provides evidence that Lo's reported percentage of cffDNA (mean ~ 3%–6%) was accepted by preeminent researchers in the field as indicative of circulating (*in vivo*) cffDNA levels and conventionally believed to be the most available for detection at the time of Lo and for years afterward. PO Resp. 37–38 (citing, *e.g.*, 2114); PO Sur-Reply 13 (citing, *e.g.*, Ex. 1030, 5; Ex. 1035, 1); Ex. 2239 ¶¶ 150, 154–155 (testimony of Dr. Van Ness about the consensus view on available cffDNA percentages and that view changing only well after the '277 patent). Petitioner does not specifically address this evidence. We are unpersuaded on this record that the POSA would have thought they should try to do better by changing Lo in the manner proposed by Petitioner.

From the foundation that a POSA would have believed a further lysis problem needed to be solved in Lo's method, a foundation we find is shaky for reasons explained above, Petitioner argues the POSA would have turned to Valli for the solution. *See, e.g.*, Pet. 52; Pet. Reply 17 ("[W]here cell lysis was a known problem (based on *Lo*), and the addition of formalin was a known solution (based on *Valli*), a POSA would have been motivated to combine these references."). This argument is also flawed and unpersuasive.

For several reasons, we question whether the POSA would have relied on Valli to the extent urged by Petitioner in its proposed combination with Lo. Valli is admittedly unrelated to DNA, cell-free DNA, or any analysis of nucleic acids. Ex. 2239 ¶ 157 ("Valli does not discuss DNA or lysis of cells"); Ex. 2294, 94:13–15 ("Q. Dr. Levy, Valli does not analyze cell-free DNA, right? A. It does not."), 171:11–18 ("Q. Can you say one way or the other whether the formaldehyde in Valli would damage cell-free DNA? . . . A. As I said, Valli did not assess cell-free DNA as any part of the scope of their research."). Valli relates to developing a "transport media for bovine blood which would improve leukocyte stability," which it studied by, *inter alia*, examining the effects of various anticoagulant combinations on bovine cell counts in whole blood shipments transported between cities in Canada over the course of several days. Ex. 1034, 252 ("The purpose of this study was to develop a transport medium for bovine blood which would provide stability during routine shipment through the mail without requiring usual precautions against Canadian climatic conditions"), 253–254 (describing experiments).

That Valli is silent on cell-free DNA, or that it was testing cell counts in transported cow's blood, does not foreclose Petitioner's combination because the claims are not limited to human blood, and the teachings on cffDNA processing and analysis come from Lo. Those differences do, however, undermine Petitioner's repeated attempts to draw parallels between Valli's and Lo's disclosures. Pet. 49 (arguing Valli and Lo's initial processing steps are "equivalent"). Lo's methods, for example and unlike Valli, collected and promptly separated the cells and plasma before freezing the plasma at -20°C for subsequent amplification and detection of cffDNA, as discussed above. There is, thus, no apparent need in Lo for cell stability over a timespan of many days as in Valli. Lo describes processing samples within hours, not days. *See* Ex. 1033, 29:5–10 (disclosing dozens of samples could be simultaneously amplified and quantified in "approximately 2 hours"); *id.* at 7:7–8 (noting "[m]aternal blood samples were processed between 1 to 3 hours following venesection"). Valli teaches that, "[i]n general, it can be stated that leukocyte counts in human blood anticoagulated with EDTA are reliable for up to 24h at room temperature and for up to 48 h at 4°C." Ex. 1034, 256. And, insofar as Valli was comparing blood transported in different anticoagulant compositions (allegedly EDTA compared to combinations with EDTA+formalin), Valli reports: "On comparing the leukocyte counts at time zero for each anticoagulant against each other, *no significant differences* were found between the anticoagulants for total white cell counts." *Id.* (emphasis added); PO Sur-Reply 17. We are unpersuaded that the POSA would have regarded Valli's concerns and results pertaining to longer-term WBC stability in certain anticoagulant compositions as especially relevant or applicable to Lo's cffDNA processing methods.[36]

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[36] Patent Owner appears to read an assertion in Petitioner's Reply as belatedly advancing a theory where Lo's method would be modified so that blood samples would be shipped away to a central lab before the samples are processed. PO Sur-Reply 14–15 (arguing that this is "new argument" and that no evidence has been presented that central labs for processing cffDNA preexisted the patent). We do not understand Petitioner to have changed its

We also agree with Patent Owner that some of Valli's results are contradicted by studies cited in Valli itself and that some of Valli's data is statistically suspect. We credit Dr. Van Ness's testimony that Valli's conclusions based on its pilot study (e.g., reporting "total white cell counts decreased with time regardless of temperature" for samples in ACD and EDTA) conflict with the non-pilot studies Valli cites. Ex. 2239 ¶ 159 (citing non-pilot study by DeLange and Lampasso (Ex. 2102, Fig. 1) reporting that WBC counts did not decrease in EDTA-treated samples at 4°C); *see also id.* ¶ 158 (testifying that a POSA "would have expected refrigerating samples to ***reduce*** degenerative cells," contrary to Valli's study). Valli also observed that WBCs somehow increased in number over time in some of its experiments, but fails to offer any explanation; this perplexing and unexplained result detracts from the weight a POSA would have given Valli's teachings. Ex. 1034, Table II; Ex. 2239 ¶¶ 158 (testimony of

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theory about the modification of Lo. Petitioner was merely pointing out that Valli was directed to compositions that allow blood samples "to be shipped to central laboratories for mass-screening." Pet. Reply 12. Had Petitioner changed its theory mid-proceeding to one that required Lo's samples be shipped to a central lab, we would find that theory new and untimely. *Intelligent Bio-Systems, Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1369 (Fed. Cir. 2016) (holding, in accord with 35 U.S.C. § 312(a)(3), that a petitioner's contentions and supporting evidence must be set forth with "particularity" in the petition itself). As explained in another case challenging the '277 patent, where such a shipment theory was made, the time associated with such shipments and delayed processing raise other material issues (e.g., exposing the cffDNA to formaldehyde for longer durations and increasing potential damage to the DNA). *See* IPR2021-00902, Paper at 28–31, 45–46.

Dr. Van Ness that Valli "neglects to address potential confounding factors" to explain how gross WBC counts "could have ***increased***"), 159 (explaining that "Valli draws its conclusions from widely variable data with 95% confidence intervals so large that none of the categories of results are statistically distinguishable"). For at least these reasons, we credit Dr. Van Ness's testimony that a POSA would have viewed Valli with skepticism, and we find a POSA would have been reluctant to so readily adopt Valli's conclusions about the alleged superiority of formalin plus EDTA as a cell stabilizer preparation. Ex. 2239 ¶¶ 158–162.

Petitioner responds that Patent Owner's argument about Valli's errors and inconsistencies conflates separate experiments in Valli. Pet. Reply 13–15. Petitioner dismisses Valli's inconsistencies and inconclusive results as being limited to Valli's pilot study[37] and Experiment I—redirecting attention to Valli's Experiment II and III, the results of which, Petitioner contends, form the basis for Valli's conclusion that formalin-EDTA is superior to EDTA alone. *Id.* at 13–14 (citing Ex. 1067, 125:13–22 (testimony of Dr. Van Ness that Valli's conclusion about anticoagulant preparations appears in a discussion of Experiments II and III)); *see also id.* at 14 n.13, 14 (asserting that Experiments II and III used EDTA alone as one of the test solutions (unlike Experiment I) and also used other methods to reduce

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[37] In a footnote, Petitioner suggests an explanation for the inconsistency in Valli's pilot about cell stability under refrigeration, noting that ACD is recommended to be stored at ambient temperature. Pet. Reply 12 n.12. Even assuming that is accurate, it does not explain Valli's inconsistent reporting on stability of cells stored in EDTA.

donor-to-donor variations). We agree with Patent Owner, however, that the inconsistencies and unreliable results elsewhere reported in Valli call into question somewhat Valli's reporting on Experiments II and III. PO Sur-Reply 17 (noting also, for example, the comparatively smaller sample sizes in Experiments II and III) (citing Ex. 1034, 255–257).

On Valli's Experiments II and III, still other questions remain. Those experiments compared blood stored in formulations described by Valli as "Titriplex I," "Titriplex IV," and EDTA. Ex. 1034, 253. Valli states that Titriplex I and Titriplex IV both contained EDTA and formalin. *Id.* at 254, 257. Patent Owner highlights, however, that Titriplex I and Titriplex IV are commercial anticoagulants that do not include EDTA or formalin. PO Resp. 40–41 (citing Ex. 2097, 32 (Titriplex IV was a known anticoagulant (DCTA, also known as CDTA)); Ex. 2098, 1 ("CDTA monohydrate" and "Titriplex® IV [are] synonyms"); Ex. 2099, 1 (nitrilotriacetic acid (NTA) and Titriplex I are synonyms)). Valli's use of the Titriplex tradenames to identify formulations with formalin and EDTA is confusing. Ex. 2239 ¶ 160 (testifying a POSA would have been skeptical of Valli's "confusing and inconsistent reporting of tested formulations"). If, nonetheless, such confusion about Valli's use of the "Titriplex" names might be addressed sufficiently by reading Valli's Table 1 as meaning that formalin and EDTA are *added* in the formulations labeled Titriplex I and Titriplex IV (Ex. 1034, 254; Ex. 1068 ¶ 16 (Dr. Levy testifying the "first and second combinations included EDTA and formalin")), that raises other issues with Valli, and Petitioner's challenge. So read, Titriplex I and IV in Valli include other compounds beyond formalin and EDTA, including other anticoagulants (e.g., CDTA or NTA). Ex. 1034, 254 (Table 1). Titriplex IV is also listed as including acetylsalicylic acid (ASA). *Id.*; *see also id.* 252 (disclosing that ASA is included "to prevent cellular clumping"), 256 (ASA "reduced the tendency of leukocytes and platelets to aggregate"). Valli was not, therefore, simply testing formalin with EDTA against the use of EDTA alone. This diminishes the conclusions about the cell-stabilizing superiority of formalin-EDTA versus EDTA alone that Valli purports to make, or that Petitioner suggests we draw from Valli. And, if Petitioner was proposing that Lo's method should be modified by wholesale substitution of the full "Titriplex I" or "Titriplex IV" formulations listed in Valli's Table 1, it never made that position clear or accounted for adding chemicals other than EDTA and formalin.

In short, even assuming Valli would have been consulted, we find the POSA would have been skeptical of the technical merits of its disclosure. We are also doubtful the POSA would have seen Valli's disclosure as particularly applicable to Lo, or as providing a solution to any need or problem in Lo's plasma-processing and cffDNA detection method.

Another important question raised in this case is whether a POSA would have been concerned with introducing formaldehyde in the sample of Lo, and formaldehyde's potential effects on DNA, especially cell-free DNA. Petitioner's declarant, Dr. Levy, admits that he did not consider any of formaldehyde's potential effects on cell-free DNA in forming his original opinions about the alleged obviousness of the claims. Ex. 2294, 113:5–114:10 (testifying "as much as I was asked to form an opinion, that was not one of the things I was asked to consider"). Dr. Levy's opinions are weakened somewhat as a result.

Patent Owner presents evidence showing that a POSA would have been aware of and concerned about detrimental effects of formaldehyde, including its effects on nucleic acids. Patent Owner cites, for example, Swenberg (Ex. 2139, 945), which reports on problems with the use of formaldehyde, including its toxic nature and formaldehyde inducing "DNA breaks" among others. PO Resp. 44–45; Ex. 2239 ¶¶ 168–169; Ex. 2139, 945 (noting chromosomal aberrations and mutations). Patent Owner also cites Srinivasan, which discloses that attempts to extract usable DNA from formaldehyde fixed tissues have only been "variably successful," explaining that "considerable evidence suggests that formaldehyde induces DNA degradation" with "few" studies reporting yield of high-molecular weight DNA. Ex. 2150, 1964 (disclosing "formaldehyde initiates DNA denaturation" and a high frequency of "sequence alteration"). Indeed, we find that Srinivasan recommends that alternatives should be used: that "the problems of formaldehyde" may be overcome by "use [of] an alternative fixative that is better suited for the preservation of nucleic acids." *Id.* at 1966; *see also* PO Resp. 44–45 (citing Ex. 2230, 2:37–43 (disclosing that formaldehyde is expensive, toxic, carcinogenic, and a danger to lab workers)). We credit Dr. Van Ness's testimony that such problems with formaldehyde would have dissuaded the POSA from choosing formaldehyde for a modification of Lo's plasma processing and cffDNA detection method—particularly in light of the fact that, on this record, there is no persuasive evidence of formaldehyde/formalin having been used in any application where cell-free DNA was the analyte prior to the ’277 patent. Ex. 2239 ¶¶ 167–170 (Dr. Levy testifying that formaldehyde’s effects on cell-free DNA “was not yet known or thoroughly vetted”).

Petitioner, in reply, argues that formaldehyde was still commonly used as a blood stabilizer despite its negative impacts. Pet. Reply 15–16 (citing Ex. 1067, 132:23–133:5). Yet that argument elides the fact that formaldehyde’s alleged known or common use was as a *cellular* fixative in applications involving, for example, cell counting (Ex. 1034) or where cellular DNA might subsequently be extracted from stabilized cells for analysis. PO Sur-Reply 1, 14, 19 (explaining, e.g., that Kiessling (Ex. 1044) related to “formalin-fixed cells”). Petitioner does not direct us to persuasive evidence suggesting, much less encouraging, that formaldehyde should be used where cell-free DNA circulating in the blood was the target. And Dr. Levy admits to not considering formaldehyde’s effects on DNA in formulating his opinions. Ex. 2299, 19:6–21.

Petitioner further responds, attempting to dismiss Swenberg partly on the basis that it relates to fixing cultured cells. Pet. Reply 16. This is a curious argument inasmuch as Petitioner’s support for the allegedly known and effective use of formaldehyde relates to *cellular* applications, using the agent to fix *cells*. *See, e.g.*, Exs. 1034 (fixing bovine blood cells), 1074 (fixing living fruit fly cells). Petitioner’s contention about Swenberg discussing longer incubation times (up to “40 days”) appears to have some merit, however, because that may have lessened concerns a POSA would have had about shorter-term exposures to formaldehyde. Ex. 2139, 950.

Petitioner's Reply fails to mention, much less directly grapple with, Srinivasan (Ex. 2150). Pet. Reply 16–17; *see generally* Ex. 1068 (Levy rebuttal declaration) (not addressing Srinivasan). We find that Srinivasan provides strong evidence that formaldehyde was known to have detrimental effects on nucleic acids. *See, e.g.*, Ex. 2150, 1964 (reporting, e.g., that formaldehyde is a widely used tissue (cellular) fixative, "[h]owever, attempts to extract usable DNA from formalin-fixed tissue for molecular biological studies have been variably successful"; noting "considerable evidence suggests that formaldehyde induces DNA degradation"; "formalin-fixed tissues exhibit a high frequency of nonreproducible sequence alteration"). Srinivasan ultimately suggests that an alternative fixative should be used: "A method to overcome the problems of formaldehyde is to use an alternative fixative that is better suited for the preservation of nucleic acids and proteins." *Id.* at 1966 (then providing discussion about alternatives to formaldehyde).

To the extent Petitioner urges that Srinivasan is encompassed by Petitioner's assertion (in a footnote of its Reply) about "non-prior art references," such assertion is unavailing. Pet. Reply 16 n.15. Neither party sought to make a showing here that the challenged claims are entitled to priority earlier than the filing of the September 11, 2003, application that matured into the '277 patent. *See supra* n.21; Section I(A) (identifying related applications). And, for argument's sake, if the claims were entitled to priority to earlier ancestral applications listed in the patent's priority chain, the earliest of which are provisional applications filed in March and May 2002, Srinivasan is explicitly a "Review" article that published in

December 2002. Ex. 2150, 1961. All of Srinivasan's seventy-four citations, including those cited in passages discussing damage to nucleic acid caused by formaldehyde, are to articles that published in 2001 or earlier. Ex. 2150, 1970–1971; PO Sur-Reply 18–19 (noting "***all cited references*** pre-dating invention"). On this record, we find that Srinivasan is informative on a POSA's perspective of the concerns with formaldehyde's use at the time of the invention, whether that be September 2003 or the filing of the aforementioned provisional applications in 2002.

In its Sur-Sur-Reply (Paper 46), Petitioner belatedly addresses Srinivasan. Indeed, Petitioner devotes nearly an entire paragraph to it—raising new and untimely argument that purports to compare Valli's formalin concentration to the concentrations of formalin reported in Srinivasan, and arguing that the combination with EDTA would help to better preserve DNA. Paper 46, 3–4 (asserting that Valli uses 1% formalin not 10% formalin, and that EDTA has a "DNase neutralizing" property as evidenced by Srinivasan).

A sur-sur-reply is not permitted as of right in IPR proceedings, and Petitioner's argument about Srinivasan in its Sur-Sur-Reply goes beyond what we authorized. *See* Ex. 3006 (Oct. 4, 2022, email from the Board setting forth the permitted scope of Petitioner's four-page sur-sur-reply). We authorized Petitioner to file a short paper, with submission of "[n]o new evidence," addressing Patent Owner's argument on the court's summary judgment order (discussed *supra*, Section II(E)(1)) and "Exhibit 2298" (an appeal brief from third-party Streck submitted in prosecution of a different patent application).[38] Petitioner did not ask, and we did not permit, Petitioner to submit new substantive argument about Srinivasan. Petitioner, nevertheless, shoehorns this argument in by claiming that Srinivasan is inconsistent with an assertion made by Patent Owner about Exhibit 2298 on whether an approach that uses cffDNA can be combined with an approach that does not employ cell-free DNA. Paper 46, 3. We see no inconsistency. Srinivasan does not specifically address cell-free DNA, much less cffDNA—it relates to using formaldehyde as a tissue (i.e., cellular) fixative and to the potential harms to nucleic acids in that context. Srinivasan does not teach that approaches using formaldehyde as a cellular fixative can or should be extended to cffDNA analysis. As Petitioner's argument about Srinivasan in the Sur-Sur-Reply was not permitted and is also unfair to Patent Owner under the circumstances (including Petitioner's earlier silence on Srinivasan, and addressing it substantively only after the evidentiary record closed), we decline to consider it. And, even if we were to consider the argument, the outcome here would not change. Although we recognize that Srinivasan notes "criteria" on recommended conditions for using formaldehyde, on balance, we find that Srinivasan highlights the damage

[38] We do not, in this proceeding, rely on Exhibit 2298 or the Board decision arising from it cited by Patent Owner. PO Sur-Reply 14–15 (citing *Ex Parte Fernando*, Appeal No. 2021-003268, 2022 WL 855866, at *12 (PTAB Mar. 21, 2022) (crediting Streck's argument that "stabilization methods suitable for stabilization of blood samples do[] not necessarily translate into suitable stabilization methods for cell-free fetal nucleic acids")). That decision was based on arguments raised by Streck in that case and the record developed there; it does not control the outcome in this IPR.

formaldehyde's use can cause to DNA and, as discussed, suggests alternatives be used. Ex. 2150, 1965–1966.

In its Reply, Petitioner also argues a POSA would understand that cell-free DNA is double stranded, and cites Orlando's disclosure that "[f]ormaldehyde does not react with free double-stranded DNA." Pet. Reply 16–17; Ex. 2294, 63:6–64:6 (testimony of Dr. Levy that "the inclination would be to think of it as double stranded"); Ex. 1073 (Orlando), 205 ("Formaldehyde does not react with free double-stranded DNA, avoiding kinetic constraints due to DNA damage."). Although that statement in Orlando, standing alone, might be seen as favoring Petitioner's position, we agree with Patent Owner that context is needed. PO Sur-Reply 17–18. The solitary statement appears in Orlando's introduction with no cited support or discussion of its basis. And, as Dr. Levy concedes, Orlando does not, in fact, relate to cell-free DNA.[39] Ex. 2299, 133:6–14 ("Q. In your understanding, did Orlando's study analyze or relate in any way to cell-free DNA? A. I don't believe that Orlando did that in this experiment."). Indeed, as explained by Patent Owner and as admitted by Dr. Levy, Orlando

[39] Patent Owner cites differences between cellular DNA and cell-free DNA as including, for example, that cell-free DNA is small, limited in quantity, and easily damaged. PO Sur-Reply 15 (Ex. 2246 ¶¶ 1, 4 (testimony of Brad Hunsley, Streck's longtime employee and Director of R&D)). Dr. Levy, on cross-examination, could provide no basis to disagree with Mr. Hunsley about those differences. Ex. 2299, 95:24–98:2. In its Sur-Sur-Reply, Petitioner asserts that Dr. Levy had not seen Mr. Hunsley's declaration before. Paper 46, 2–3. Petitioner did not, however, move to exclude the declaration or Dr. Levy's further testimony about it, or seek authorization to move to strike Patent Owner's contentions about the same.

relates to using formaldehyde for *in vivo* fixation of fruit fly *cells*, from which chromatin (a *cellular* DNA/protein complex) is later extracted and analyzed. PO Sur-Reply 18 (noting Orlando's method involved "cross-linking cells with formaldehyde, halting the cross-linking reaction, and washing the cells (thus removing any extracellular components) before analysis") (citing Ex. 1073, Title, 206, Fig. 1; Ex. 2299, 134:7–15, 135:9–25, 137:11–138:20). This left Dr. Levy to admit that he relied on the "statement" about formaldehyde and "free DNA" divorced from the actual context of Orlando. Ex. 2299, 139:18–140:3 ("My use of Orlando was to show a published reference with that statement and to back up that statement that I've made in my declaration."). Considering the full record, Petitioner does not persuade us that a POSA would expect, from Orlando's unrelated methods and unsubstantiated single sentence about "free" DNA, that formaldehyde would neither damage nor affect cell-free fetal DNA.

Lastly, Petitioner asserts that Patent Owner relies on vague, unknown effects about formaldehyde that do not undermine Petitioner's challenge. Pet. Reply 17. We disagree because the evidence shows that it was known that formaldehyde can damage DNA, even in cellular applications. To the extent the effects on cell-free DNA are characterized as "unknown," that is partly true because, on this record, no one had used formaldehyde in the analysis of cell-free DNA prior to the '277 patent, and only underscores the lack of obviousness shown here. Petitioner asserts that Patent Owner's concerns about formaldehyde effects are not expressed in the patent, nor have any novel techniques been described to combat them. But we remain unpersuaded on this record that this suggests the challenged claims would have been obvious. The point of novelty for the challenged claims was the addition of a lysis-inhibiting agent, formaldehyde, in particular, to cffDNA processing and analysis. *See supra* Section I(E). And the patent includes detailed working examples showing how that can be done, with accompanying results. *See, e.g.*, Ex. 1001, 89:1–91:50 (Example 4).

Altogether, we find that the evidence of record supports that formaldehyde was known to have detrimental effects on nucleic acids. We also find that, notwithstanding the known use of formaldehyde as a stabilizer for cells, a POSA would have had significant and unresolved concerns about expanding formaldehyde's use to applications where cell-free fetal DNA was the analyte as such use could damage the cffDNA in the sample itself, frustrating its detection and analysis. And we credit the testimony of Dr. Van Ness that such concerns would have dissuaded a POSA from modifying the Lo method with Valli in the manner proposed by Petitioner.

For the above reasons, considering the argument and evidence presented through trial, Petitioner does not persuade us that a POSA would have been motivated to combine Lo and Valli to arrive at the subject matter of claims 55 and 81. Petitioner's challenge to the dependent claims under Ground 3 relies on its analysis for claims 55 and 81 analysis (Pet. 55–65), and Petitioner does not argue or show that its challenge to the dependent claims makes up for deficiencies we have noted above. Pet. Reply 10–17 (same responsive argument for all claims). Thus, we conclude that Petitioner has not proved that claims 56–63, 66–69, 80, 82–91, 94, 96, 126–130, 132, 133 are unpatentable based on the combination of Lo and Valli. *In re Fritch*, 972 F.2d 1260, 1266 (Fed. Cir. 1992) ("[D]ependent claims are nonobvious if the independent claims from which they depend are nonobvious.”).

Petitioner challenges claim 95 based on the combination of Lo, Valli, and Marx (Ground 4). Pet. 65–66; Ex. 1056 ¶¶ 152–153. Petitioner’s theory is based on combining Lo and Valli, as discussed above, with a further addition of no-brake centrifugation from Marx (offering the same reasons for adding Marx as in Ground 2, avoiding disturbance of the buffy coat). Pet. 65–66. Patent Owner does not present any argument on Ground 4, separate from the argument about Lo and Valli, which we addressed immediately above. PO Resp. 36 (arguing “Grounds 3–4” together). Petitioner’s showing for Ground 4 does not address or remedy the deficiencies noted above with Ground 3 and the combination of Lo and Valli. Accordingly, Petitioner also has not shown by a preponderance of the evidence that claim 95 is unpatentable over Lo, Valli, and Marx.

III. MOTIONS FOR PROTECTIVE ORDER AND TO SEAL

Patent Owner moves for entry of a stipulated protective order and for an order sealing portions of the Patent Owner Response (Paper 22), all of Exhibits 2170–2173, and portions of Exhibit 2080. *See* Paper 23 (attaching stipulated protective order as Appendix A). That motion is unopposed.

A party may move to seal confidential information including, *inter alia*, sensitive commercial information. Consolidated Patent Office Trial Practice Guide, 19 (Nov. 2019); 37 C.F.R. § 42.54. It is the movant’s burden to show good cause for sealing such information, and we balance the party’s asserted need for confidentiality with the strong public interest in open proceedings. *Argentum Pharms. LLC v. Alcon Research, Ltd.*, IPR2017-01053, Paper 27 at 4 (PTAB Jan. 19, 2018) (informative).

Patent Owner provides a sufficient explanation for sealing the relevant portions of the Patent Owner Response and the identified exhibits. Exhibits 2170–2173 are license or settlement agreements between Patent Owner and third parties setting forth, for example, payment terms and sales data. Patent Owner states that it produced those agreements with the permission of third parties on the condition that the agreements remain sealed. Paper 23, 3–4. The subject portions of the Patent Owner Response and Exhibit 2080 include discussions about those agreements. Patent Owner has also provided public redacted versions of Exhibit 2080 and the Patent Owner Response (Paper 24) so the record may remain clear and reasonably open.

Patent Owner has established good cause for sealing Exhibits 2170–2173, portions of Ex. 2080, and the Patent Owner Response. The stipulated Protective Order includes minor changes from our default language. That Protective Order, Appendix A to Paper 23, is entered.

Petitioner also filed a Motion to Seal (Paper 31). Petitioner contends that portions of its Reply (Paper 34) and portions of Exhibit 1066 (deposition transcript of Paul K. Meyer) should be sealed because those papers include information that Patent Owner considers contain confidential business information (e.g., licensing practices). Paper 31, 1. Petitioner has also filed public redacted versions of the Petitioner Reply (Paper 35) and Exhibit 1066. For the same reasons discussed above regarding Patent

Owner's Motion to Seal, Petitioner has shown good cause for granting its motion. Petitioner's Reply (Paper 34) and Exhibit 1066 are sealed.

IV. CONCLUSION

On this record, Petitioner has not shown by a preponderance of the evidence that any of the challenged claims are unpatentable based on grounds advanced in the Petition.

In Summary:

Claims	**35 U.S.C. §**	**Reference(s)/Basis**	**Claims Shown Unpatentable**	**Claims Not shown Unpatentable**
55–59, 61–63, 66–69, 80–89, 94, 96, 126–130	102(a), (e)	Landes		55–59, 61–63, 66–69, 80–89, 94, 96, 126–130
95	103(a)	Landes, Marx		95
55–63, 66–69, 80–91, 94, 96, 126–130, 132, 133	103(a)	Lo, Valli		55–63, 66–69, 80–91, 94, 96, 126–130, 132, 133
95		Lo, Valli, Marx		95
Overall Outcome				55–63, 66–69, 80–91, 94–96, 126–130, 132, 133

V. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that the Petitioner has not proved by a preponderance of the evidence that claims 55–63, 66–69, 80–91, 94–96, 126–130, 132, and 133 are unpatentable;

FURTHER ORDERED that Patent Owner's Motion for Entry of Stipulated Protective Order (Appendix A to Paper 22) and to Seal is *granted*;

FURTHER ORDERED that Petitioner's Motion to Seal (Paper 31) is *granted*; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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